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Supporting Information

High-Capacity Adsorption and Thermal Destruction Adaptable to PFAS Chain Length via Crystal-to-Crystal Transformation of Zirconium-Organic Framework

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These authors contributed equally to this work.

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Materials and methods

Chemicals. Zirconium(IV) chloride, benzoic acid, 4,4',4",4'"-(4,4'-(1,4-Phenylene)bis(pyridine-6,4,2-triyl))tetrabenzoic acid (PTA), *N*,*N*-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), hydrochloric acid, sodium hydroxide, humic acid, NaCl, $NaNO_3$, Na_2SO_4, $Al_2(SO_4)_3$, $CaSO_4$, $MgSO_4$, $CuSO_4$, KI, $FeCl_3$, LiCl, $ZnCl_2$, $MnSO_4$, $Co(NO_3)_2$, NaBr, trifluoroacetic acid (TFA), perfluoropropionic acid (PFPrA), perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUdA,), perfluorododecanoic acid (PFDoA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), perfluorohexadecanoic acid (PFHxDA), perfluorooctadecanoic acid (PFODA), perfluorobutanesulfonic acid (PFBS), perfluorohexanesulfonic acid (PFHxS), perfluorooctane sulfonate (PFOS), perfluorodecanesulfonic acid (PFDS), hexafluoropropylene oxide dimer acid (HFPO-DA), and 6:2 fluorotelomer sulfonic acid (6:2 FTS) were purchased from commercial sources and were used without further purification.

Powder X-ray diffraction (PXRD). PXRD patterns were collected on a Rigaku ULTIMA IV powder diffractometer with a Cu microfocus tube at 40 kV, 40 mA, and 1600 W.

Single-crystal X-ray crystallography (SCXRD). The single crystal of **Zr-PTA1** was directly transferred from the mother liquor to the oil, and then mounted onto a loop for SCXRD measurements. The data were collected on a Bruker D8-Venture diffractometer equipped with Cu microfocus tubes (λ = 1.54178 Å). The crystal was kept at 100.00 K during data collection. Using Olex2,[1] the structure was solved with the SHELXT[2] structure solution program via Intrinsic Phasing and refined with the XL[3] refinement package using Least Squares minimisation. Crystal data are summarized in Table S1 and the raw data can be obtained free of charge from the Cambridge Crystallographic Data Centre (CCDC) under deposition number 2453981.

Single crystal electron diffractometer. The single-crystal sample of **Zr-PTA2** was dispersed in acetone and transferred onto ultrathin carbon-coated copper grids. The specimen was immobilized using a cryogenic transfer system (liquid nitrogen environment, 100 K) and loaded into a JEM-2300ED (JEOL Ltd., Japan). Electron diffraction data were acquired at an accelerating voltage of 200 kV. The instrument was equipped with a horizontally movable 4-axis goniometer and a HyPix-ED detector (effective area: 77.5 mm × 36.5 mm). A universal cryo-transfer holder (Gatan 689.STP) was employed for sample handling. Data acquisition was controlled by CrysAlisPro for XtaLAB Synergy-ED software. Three-dimensional diffraction data were collected via the continuous rotation method (−60° to +60°, step size 1°) after crystal screening by nano-beam electron diffraction (NBED). Crystallographic parameters are summarized in Table S2, and structural data have been deposited with CCDC under deposition number 2453980.

Thermogravimetric analysis (TGA). TGA was conducted on a Mettler-Toledo TGA/DSC 1 under a nitrogen (N_2) atmosphere with a ramp rate of 10 °C/min from room temperature to 800 °C.

Scanning electron microscopy (SEM). SEM was carried out using a TESCAN MIRA LMS scanning electron microscope. The samples were dispersed on conductive adhesive, coated with gold for 45 s using a Quorum SC7620 sputter coater (current: 10 mA).

Fourier transform infrared spectroscopy (FTIR). FTIR data were collected on a Thermo Nicolet iS50 spectrophotometer with a scanning range of 400–4000 cm^{-1}.

N_2 sorption measurement. N_2 adsorption-desorption isotherms were measured on a Micromeritics ASAP 2020 system at 77 K. The synthesized **Zr-PTA1** sample was soaked in

chromatographic-grade acetone for 72 h (with the solvent refreshed three times daily) to exchange the solvents within the pores. The resulting sample was then freeze-dried for 24 h under vacuum. Prior to measurement, **Zr-PTA1** was further activated using a Micromeritics Smart VacPrep by heating at 373 K for 10 h under ultra-high vacuum using a diaphragm and turbo pumping system. The specific surface area was calculated using the Brunauer-Emmett-Teller (BET) model.

Zeta potential measurement. Zeta potential was measured on an Anton Paar Litesizer 500 at 25 ± 0.5 °C using Phase Analysis Light Scattering (M3-PALS).

Contact angle measurement. Contact angles wAere measured using an SDC 350KS instrument (Kunshan Shengding, China) via the sessile drop method with 10 μL deionized water. The angle was calculated by fitting with the Young-Laplace equation.

Transmission electron microscopy (TEM). Samples were dispersed in ethanol by ultrasonication, deposited onto copper grids, and air-dried. TEM imaging was performed on a JEOL JEM-F200 microscope at 200 kV, equipped with a JED-2300T energy dispersive X-ray spectroscopy (EDS) detector.

Nuclear magnetic resonance (NMR) spectroscopy. ^{1}H NMR spectra were recorded on a Bruker AVANCE NEO 400 MHz spectrometer equipped with a 5 mm Z-gradient multinuclear probe at room temperature.

Inductively coupled plasma mass spectrometry (ICP-MS). The background elemental compositions in real water matrices and the precise concentration of leached metal ions were quantified using an Agilent 7800 ICP-MS spectrometer. The optimized instrumental operating parameters were set as follows: RF power of 1550 W, nebulizer gas flow of 1.00 L/min, auxiliary gas flow of 1.00 L/min, peristaltic pump rate of 20 r/min, and a sample flush time of 40 s.

X-ray photoelectron spectroscopy (XPS). XPS analysis was performed using a Thermo Scientific K-Alpha spectrometer (Thermo Fisher Scientific) under an ultrahigh vacuum (base pressure: 5×10^{-9} mbar). Monochromatic Al Kα radiation (1486.6 eV, 12 kV accelerating voltage, 6 mA beam current) was employed as the excitation source. Binding energies were referenced to the adventitious carbon C 1s peak at 284.8 eV for charge correction. Peak deconvolution and quantitative analysis were performed using Avantage 5.9931 software.

X-ray absorption fine structure (XAFS). Zr K-edge analysis was performed at the BL11B beamline of the Shanghai Synchrotron Radiation Facility (SSRF, Shanghai, China) using a Si (111) double-crystal monochromator. Prior to analysis, samples were pressed into thin sheets of 1 cm diameter and sealed with Kapton tape. EXAFS spectra were collected in transmission mode. Data processing and EXAFS fitting were performed using the Athena and Artemis programs of the Demeter software package[4], which employs the FEFF6 code.[5]

Synthesis of Zr-PTA1. Zirconium (IV) chloride (31 mg, 0.13 mmol), benzoic acid (1440 mg, 11.8 mmol), and the ligand PTA (22 mg, 0.05 mmol) were placed in a glass vial. Subsequently, 4.5 mL of DMF was added, and the mixture was sonicated for 20 min to ensure complete dissolution. The homogeneous solution was transferred to a 25 mL Teflon-lined stainless-steel autoclave and heated statically at 120 °C for 96 h. After cooling naturally to room temperature, the white crystalline product was collected via centrifugation and washed repeatedly with fresh DMF (5 × 5 mL) to remove unreacted precursors. The final product was freeze-dried for 24 h to yield **Zr-PTA1** as a white powder. The synthesis methodology was established with reference to the studies conducted by Wang et al.[6] and Joseph et al.[7]

Synthesis of Zr-PTA2. Freshly synthesized **Zr-PTA1** crystals (500 mg) were dispersed in 60 mL of DMF under a nitrogen atmosphere. Deionized water (0.7 mL) and concentrated

hydrochloric acid (37%, 12 M, 1.5 mL) were sequentially added. The mixture was stirred continuously at 100 °C in an oil bath for 24 h at 400 rpm. The resulting white precipitate was isolated by centrifugation, washed with DMF (5 × 5 mL), and freeze-dried for 24 h to obtain the porous **Zr-PTA2** material.

Adsorption experiments for perfluoroalkyl substances (PFAS). The synthesized **Zr-PTA1** and **Zr-PTA2** materials were pre-treated by washing with DMF (3 × 5 mL) and acetone (3 × 5 mL), followed by freeze-drying for 12 h. Batch adsorption experiments were conducted by dispersing the materials in aqueous solutions containing PFAS at varying concentrations: PFOA: 0–4000 mg/L, PFOS: 0–3000 mg/L, PFBA: 0–500 mg/L, PFBS: 0–200 mg/L. A mass-to-volume ratio of 0.4 g/L (e.g., 4 mg MOF in 10 mL solution) was maintained. The mixtures were shaken at 25 °C and 150 rpm for 24 h to reach adsorption equilibrium. After filtration through a 0.22 μm aqueous phase membrane filter, the residual PFAS concentrations in the supernatants were quantified via ultra performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS).

The removal efficiency of PFOA was calculated by eq S1:

$$\mathrm{PFOA}_{\mathrm{Removal}} = \frac{C_0 - C_t}{C_0} \times 100 \qquad \text{eq S1}$$

where C_0 (mg/L) and C_t (mg/L) are the initial and residual concentration of PFOA in the stock solution and filtrate, respectively.

The adsorption capacity was determined by eq S2:

$$Q_t = \frac{C_0 - C_t}{C_A} \times 1000 \qquad \text{eq S2}$$

where Q_t (mg/g) is the PFAS uptake capacity by the MOF adsorbent at time t (h), and C_A (mg/L) is the concentration of adsorbent.

Langmuir adsorption fit was generated by Non-linear Least Square Regression in eq S3:

$$\frac{1}{Q_e} = \frac{1}{Q_m} + \frac{1}{C_e \times Q_m \times K_L} \qquad \text{eq S3}$$

Where Q_e (mg/g) is the amount of PFAS adsorbed at equilibrium, Q_m (mg/g) is the theoretical maximum adsorption capacity of adsorbent at equilibrium, C_e (mg/L) is the equilibrium concentration of the PFAS in the solution following adsorption, and K_L (L/mg) is a Langmuir-adsorption-affinity constant.

Adsorption kinetics experiments. **Zr-PTA1** (0.4 g/L) was dispersed in an aqueous solution containing 3000 mg/L PFOA, 500 mg/L PFOS, 200 mg/L PFBA, or 200 mg/L PFBS, respectively. The mixture was shaken at 25 °C and 150 rpm. Samples (1 mL) were collected at 0, 1, 5, 15, 30 min; and 1, 2, 4, 7, 12, 24 h, filtered through a 0.22 μm aqueous membrane, and the filtrate was analyzed for PFAS concentration. **Zr-PTA2** (0.4 g/L) was dispersed in solutions of PFBA or PFBS (each at 400 mg/L), with sampling and analysis performed under identical conditions. All experiments were conducted in triplicate or duplicate, and data are presented as mean ± standard deviation.

The pseudo-first-order and pseudo-second-order kinetic models for each adsorbent were linearized as follows:

$$Q_t = Q_e(1 - exp(-k_1 t)) \qquad \text{eq S4}$$

where k_1 (h^{-1}) is the pseudo-first-order rate constant.

The pseudo-second-order kinetic model for each adsorbent was linearized as follows:

$$\frac{t}{Q_t} = \frac{t}{Q_e} + \frac{1}{k_2 Q_e^2} \quad \text{eq S5}$$

where k_2 (g/(mg·h)) is the pseudo-second-order rate constant.

Flow-through column experiments. These experiments employed a glass chromatographic column with an inner diameter (*d*) of 1.1 cm and a total height of ≤10 cm, featuring a triple-layer structure: a compacted 0.3 cm medical degreased cotton base layer for support, a 4.2 cm (*H*) middle layer filled with 1.95 g (*m*) of MOFs material (density 1.63 g/cm^3 (ρ)), and a 0.3 cm medical degreased cotton top layer to prevent material loss. After column packing, deionized water was percolated at 0.2 mL/min for 24 h for pre-saturation. During the adsorption phase, the influent was switched to a 10 μg/L PFOA solution (prepared with deionized water) while maintaining a flow rate of 0.2 mL/min. Effluent samples were collected continuously over 10 pore volumes from the onset of PFOA injection. A control experiment was conducted using an identical column fully packed with degreased cotton. The entire experiment was performed under controlled temperature conditions at 25 ± 1 °C.

Recycling experiments. **Zr-PTA1** was regenerated from PFOA@**Zr-PTA1** samples by immersing in methanol for ~30 h at room temperature. The recycling experiment was conducted by repeating the PFOA adsorption and desorption experiments successively. The same regeneration and recycling procedures were applied to the PFBA-adsorbed **Zr-PTA2** material (PFBA@**Zr-PTA2**).

PFAS degradation experiments. PFOA-loaded composite (PFOA@**Zr-PTA1**, PFAS loading ~2.5 g/g MOF) was dispersed in a DMSO/H_2O (8:1, v/v) solvent, with an initial PFOA concentration of 500 mg/L. The mixture was transferred to a 20 mL high-temperature-resistant glass reactor and treated with 15 equivalents of NaOH (relative to PFOA), and then stirred magnetically (400 rpm) at 90 °C for 45 h. To prevent structural damage to **Zr-PTA1** from excessive alkalinity, the NaOH was added in 20 equal aliquots over a specified period. Samples (1 mL) were collected at 0, 1, 2, 6, 12, 24, 36, and 45 h, and immediately quenched in an ice bath. The samples were analyzed for the following: PFCAs (PFPeA, PFHxA, PFHpA, and PFBA) via HPLC-MS; and for fluoride (F^-), TFA, PFPrA, and small organic acids (glycolate, formate, tartronate, carbonate, and oxalate) via ion chromatography.

PFAS analytical methods. Quantification was performed using UPLC-MS/MS (Waters APGC/UPLC Xevo TQ-S). Chromatographic separation employed an ACQUITY UPLC BEH C18 column (2.1 × 50 mm, 1.7 μm) at 40 °C, with a mobile phase of 2 mM aqueous ammonium acetate (A) and methanol (B). The HPLC gradient program (mobile phase B composition and flow rate) was performed as detailed in Table S10, with an injection volume of 10 μL. MS detection used ESI-negative mode with parameters: drying gas 350 °C, flow 800 L/h, nebulizer 7.0 bar, capillary voltage 3.12 kV. Multiple Reaction Monitoring (MRM) transitions for PFAS were monitored as summarized in Table S11, with collision energies optimized for each analyte. Calibration curves (1–100 μg/L) were established using external standards. Quality controls included blank samples, spiked recoveries (80–120%), and precision (RSD ≤15%). Polypropylene consumables were used to avoid fluoropolymer interference.

Analysis of degradation products. Ion chromatographic separation was performed on a Dionex ICS-5000 system (Thermo Fisher Scientific) equipped with two distinct analytical columns. For the determination of trifluoroacetate, carbonate (CO_3^{2-}), tartronate ($C_3H_3O_5^{2-}$), oxalate ($C_2O_4^{2-}$), and perfluoropropionate, an IonPac AS11-HC anion-exchange column (4 × 250 mm) was employed under isocratic elution with 20 mM KOH at a flow rate of 15.0 mL/min. Fluoride (F^-), glycolate ($C_2H_3O_3^-$), and formate ($HCOO^-$) were quantified using a SH-AC-23 anion column (4 × 250 mm) with 20 mM KOH eluent at 5.0 mL/min. Prior to injection,

aqueous samples were filtered through 0.22 μm aqueous membrane to remove particulate matter. The separation efficiency and elution profiles of target anions were validated through representative ion chromatograms.

Theoretical calculation method. All density functional theory (DFT) calculations were performed using the Vienna Ab-initio Simulation Package (VASP).[8, 9] The exchange-correlation effects were treated with the Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) method.[10, 11] The core-valence interactions were accounted for by the projected augmented wave (PAW) method.[12] An energy cutoff of 500 eV was used in this work. A plane-wave energy cutoff of 500 eV was employed. The Brillouin zone was sampled using a Γ-centered k-point mesh of 1 × 1 × 1. Structural optimization was considered converged when the energy and force criteria were below 1.0×10^{-5} eV and 0.03 eV/Å, respectively.

Supplementary figures

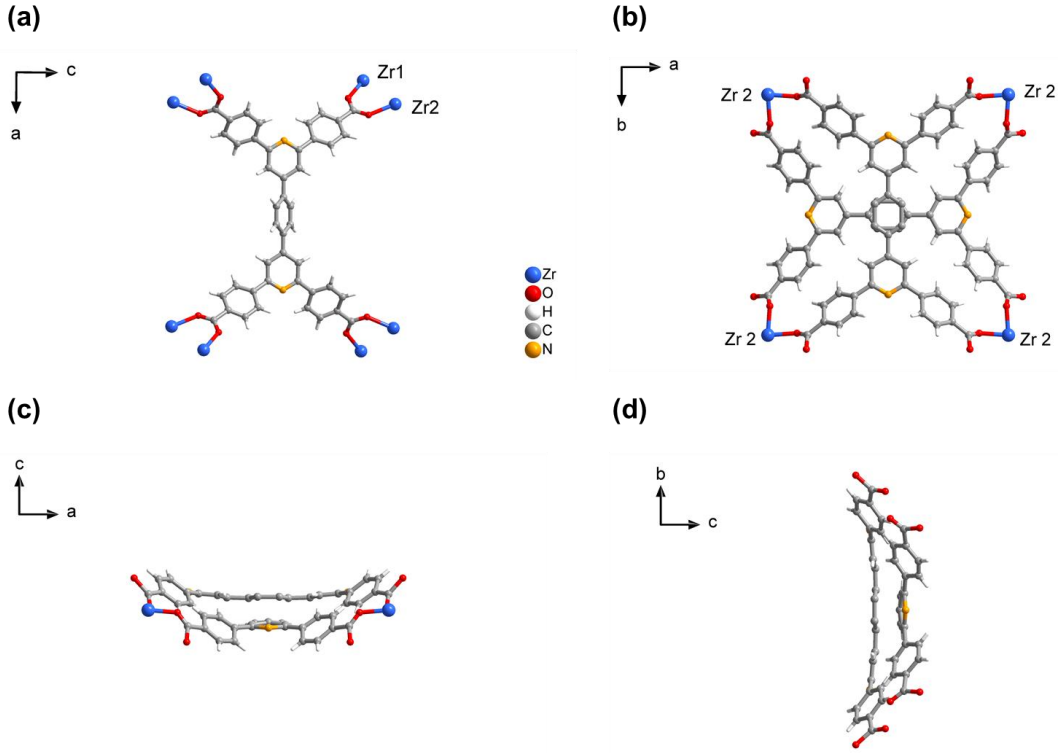


Figure S1. (a) PTA and (b-d) disordered PTA in **Zr-PTA1**.

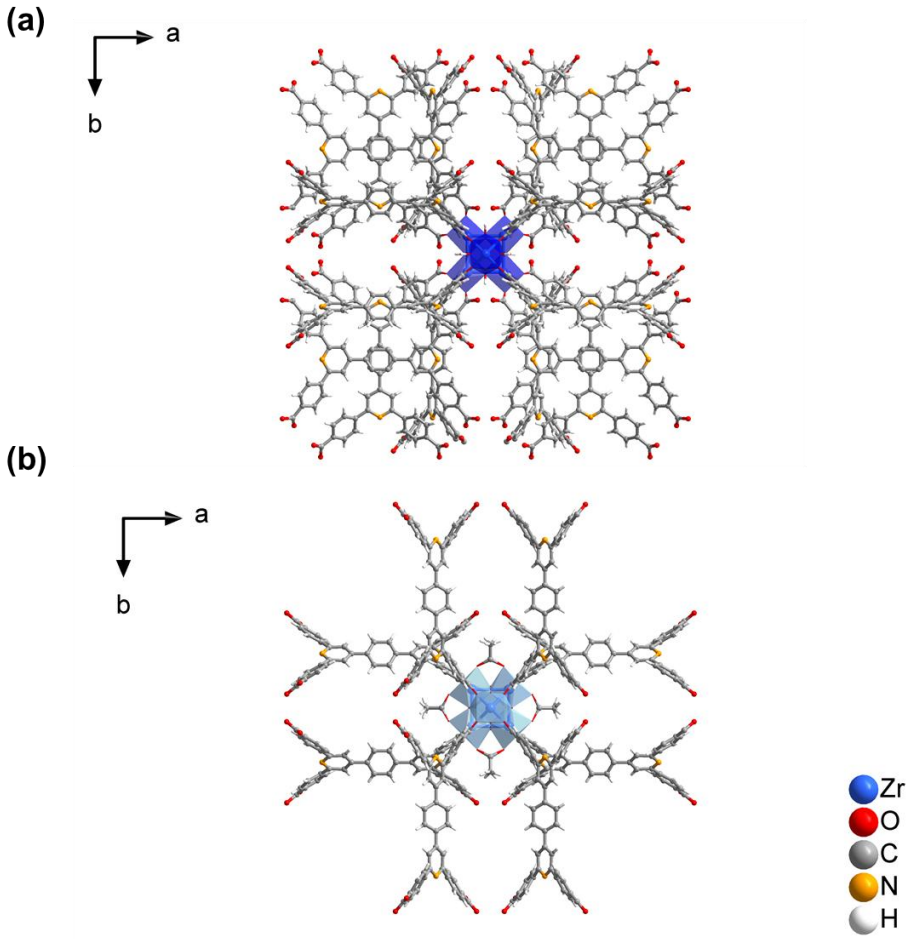


Figure S2. (a) The coordination mode between Zr clusters and ligands of **Zr-PTA1**. (b) The coordination mode between Zr clusters and ligands of **Zr-PTA2**.

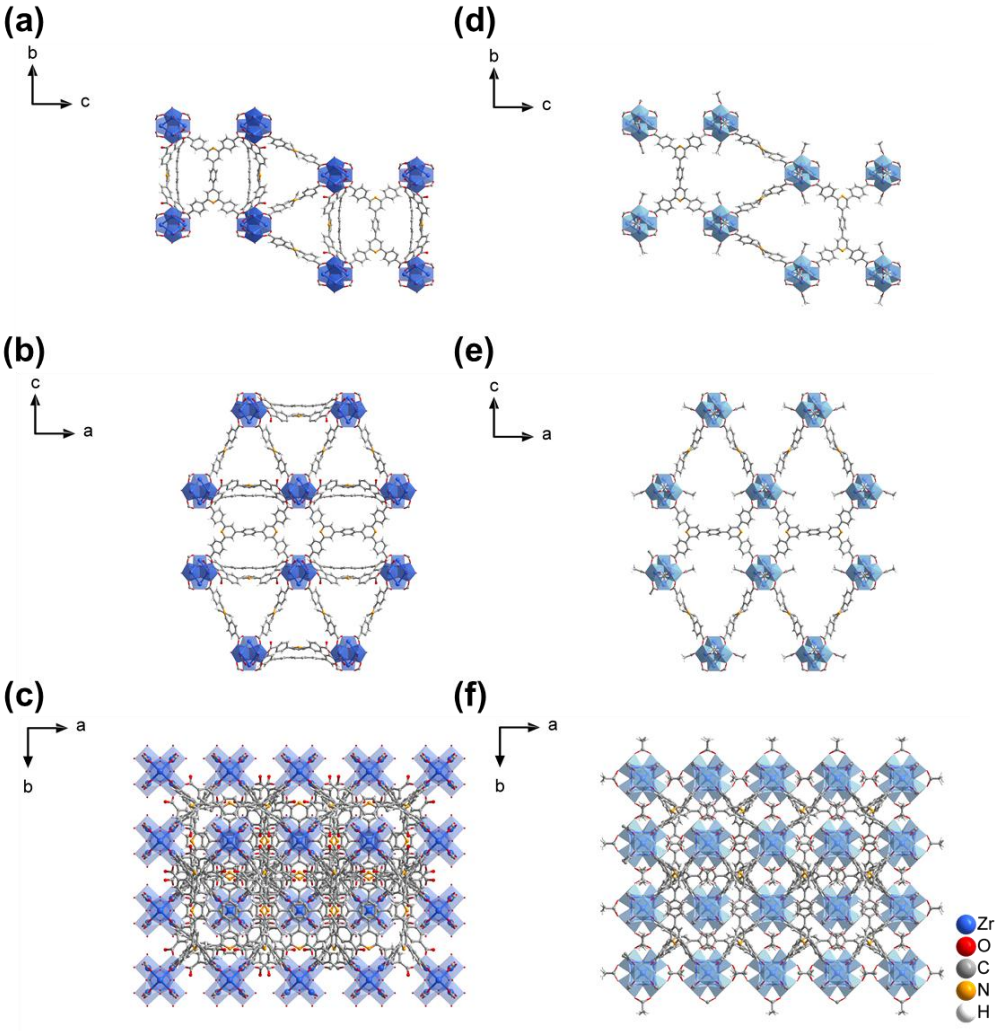


Figure S3. (a–c) Structure views of **Zr-PTA1**. (d–f) Structure views of **Zr-PTA2**.

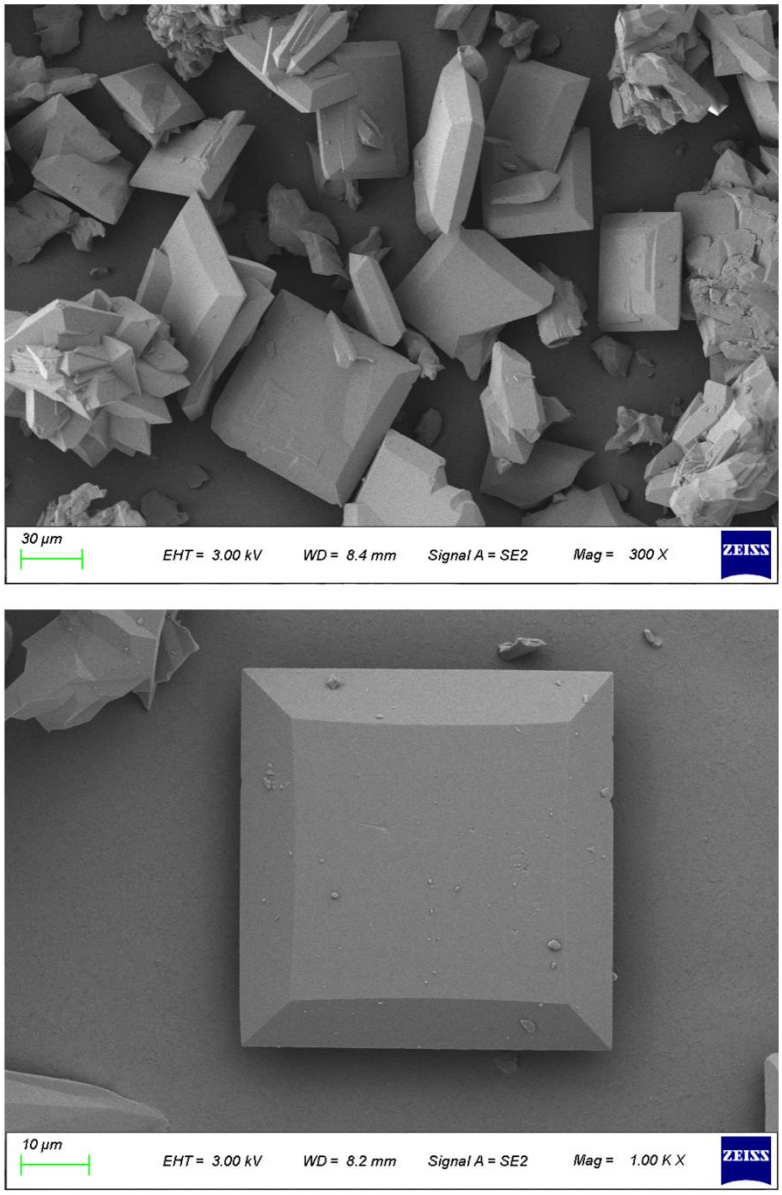


Figure S4. SEM images of **Zr-PTA1**.

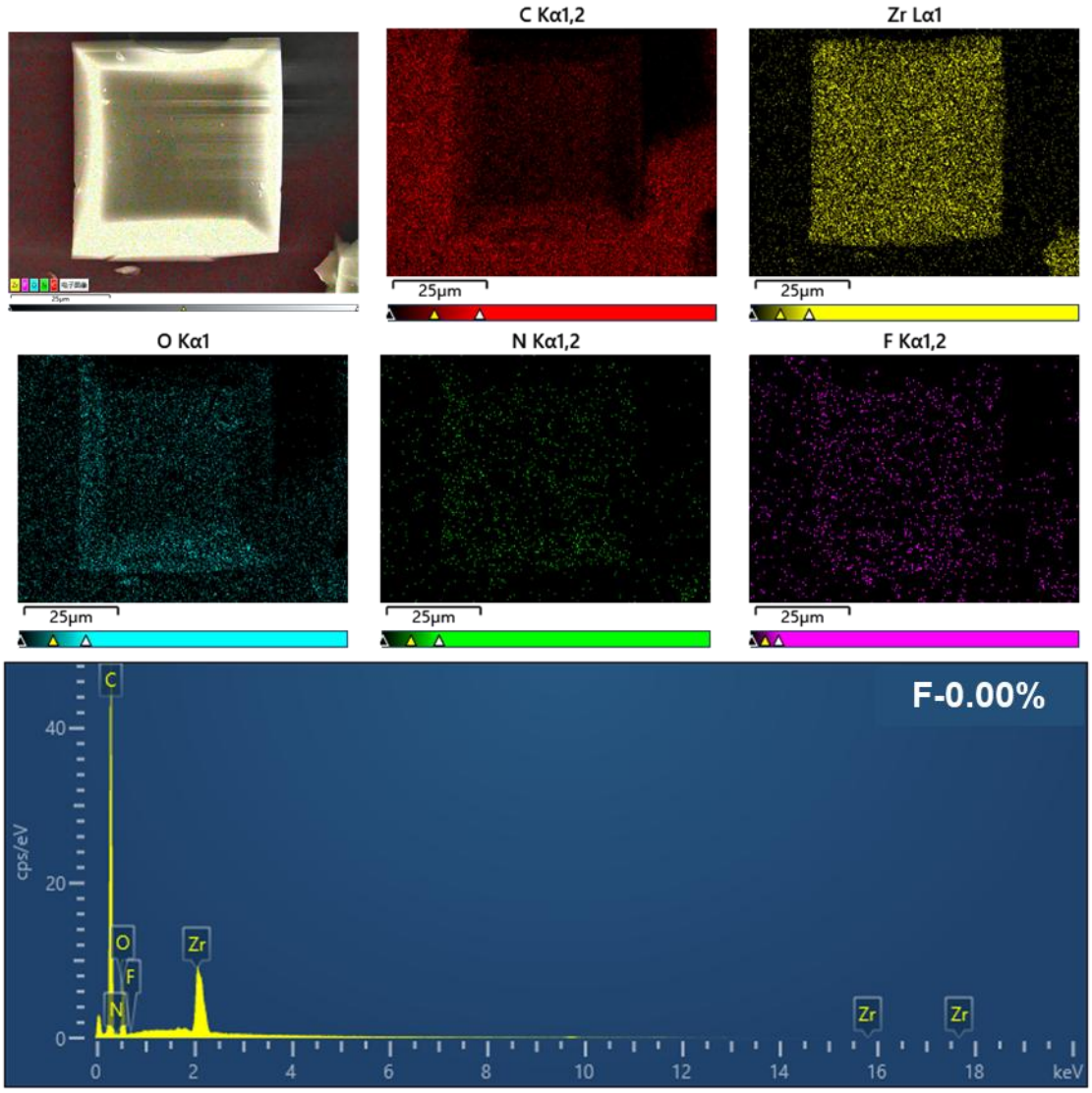


Figure S5. EDS mapping images of **Zr-PTA1**.

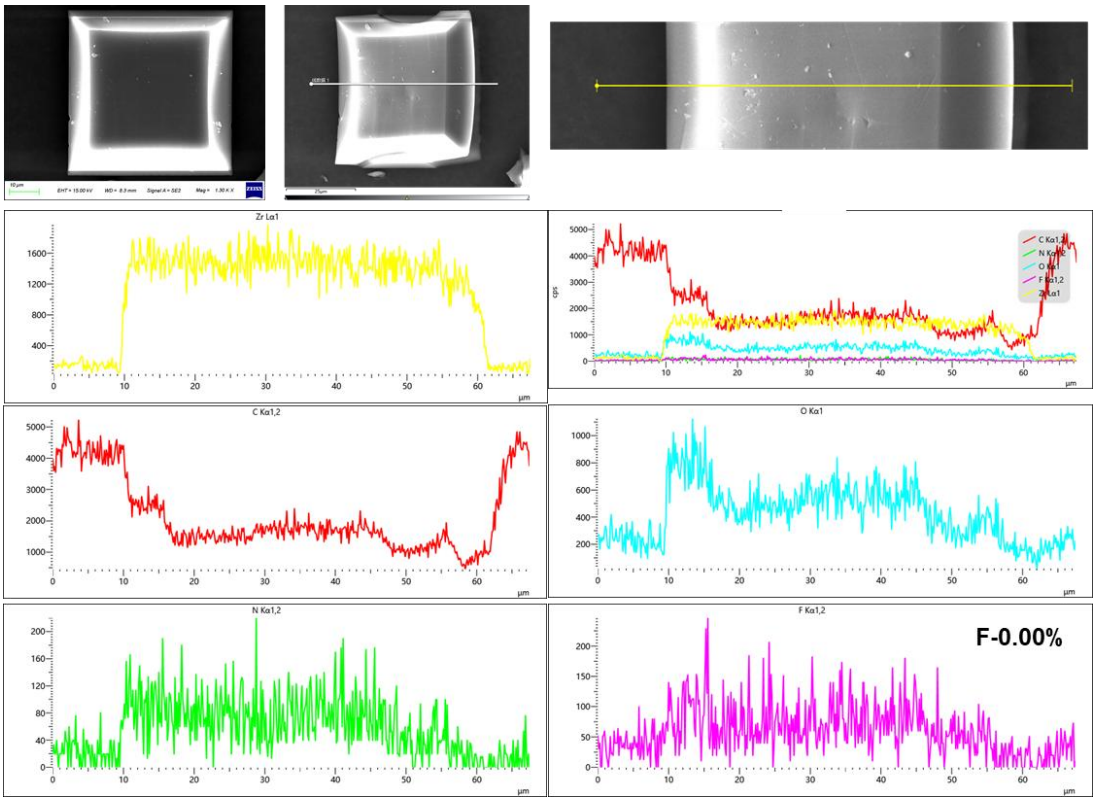


Figure S6. EDS line spectra of **Zr-PTA1**.

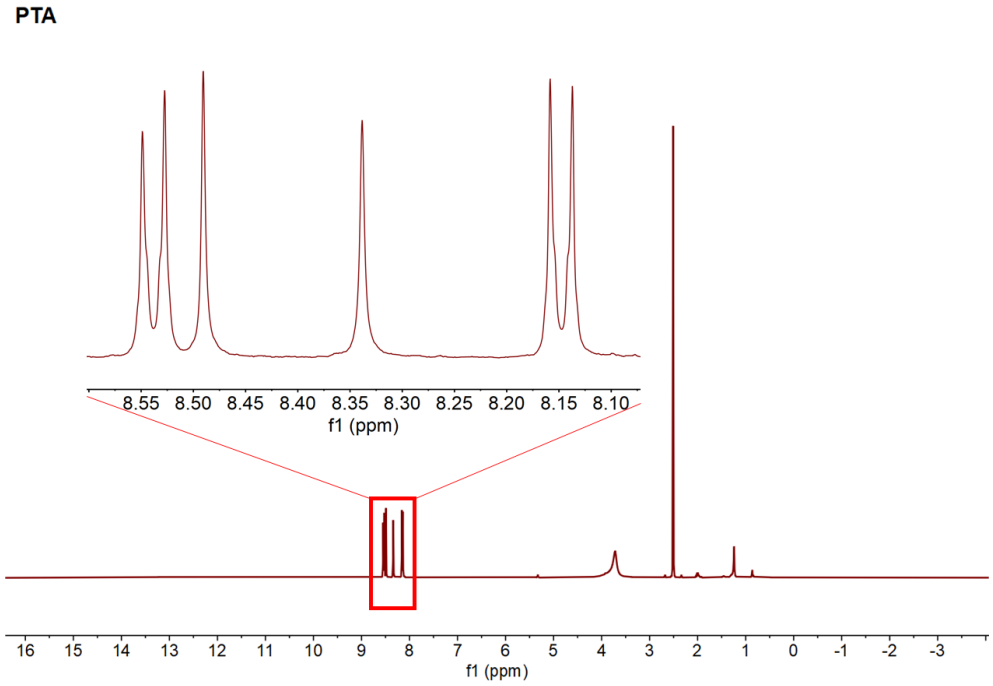


Figure S7. ^{1}H NMR spectra of the PTA ligand (DMSO-d_6, 400 MHz).

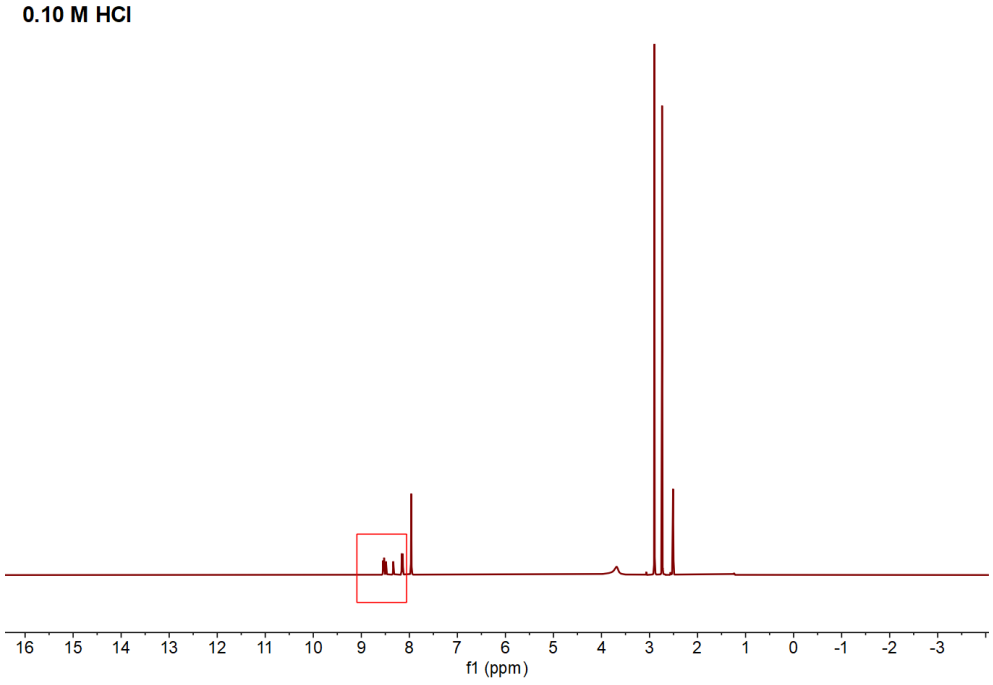


Figure S8. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.10 M HCl for 24 h (DMSO-d_6, 400 MHz).

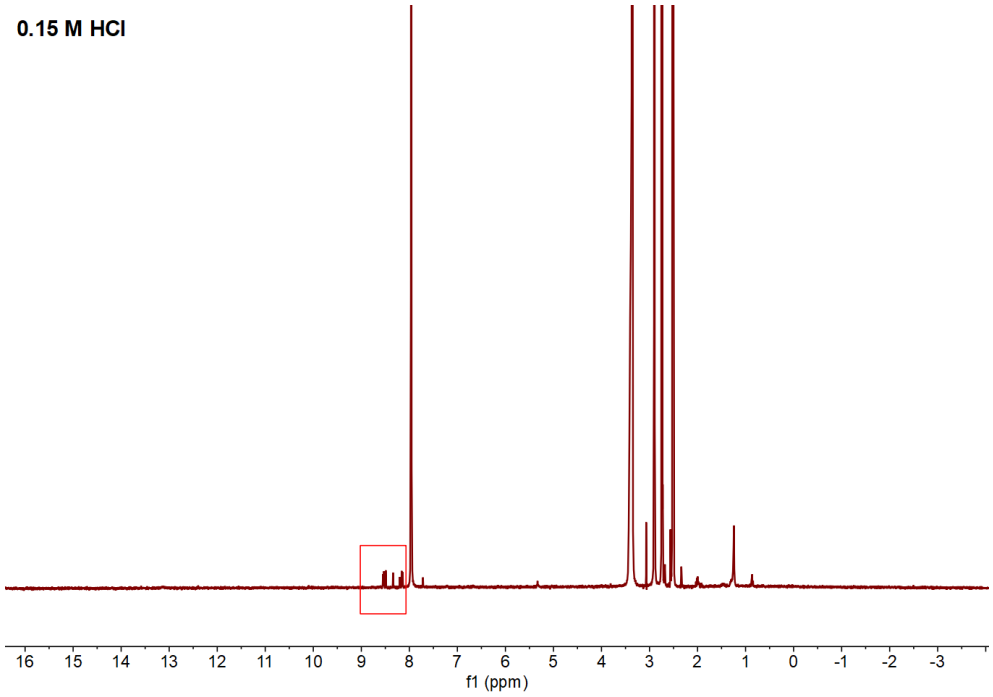


Figure S9. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.15 M HCl for 24 h (DMSO-d_6, 400 MHz).

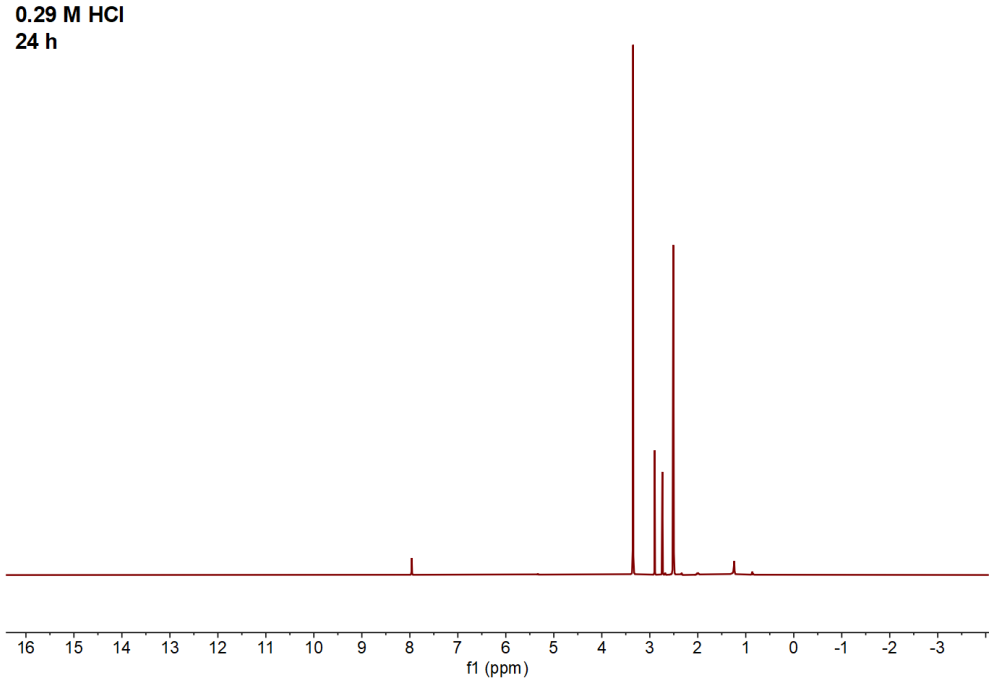


Figure S10. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.29 M HCl for 24 h (DMSO-d_6, 400 MHz).

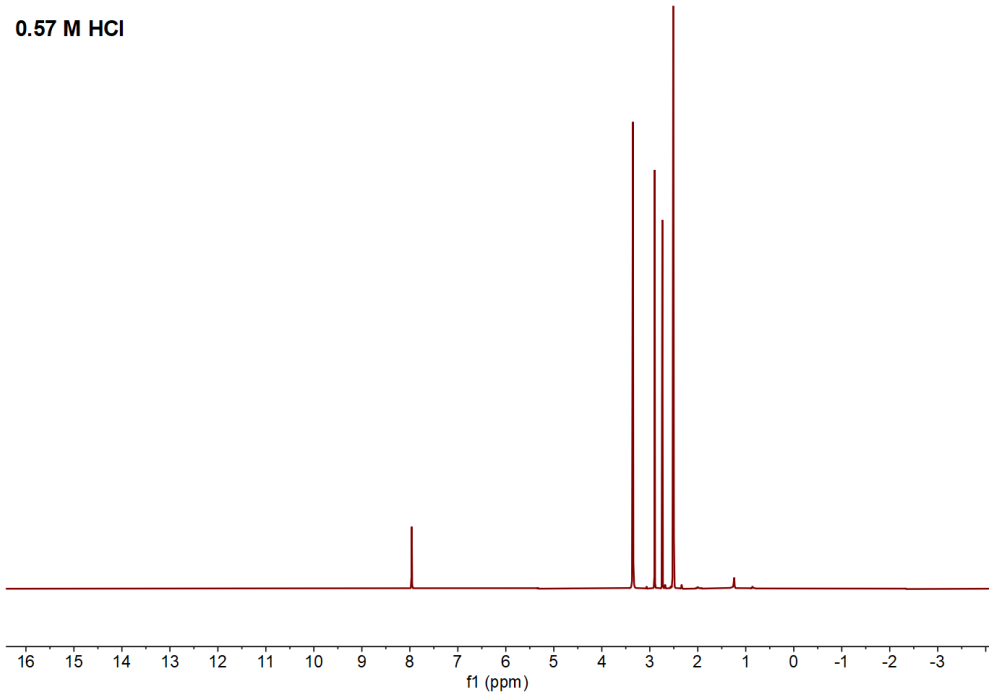


Figure S11. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.57 M HCl for 24 h (DMSO-d_6, 400 MHz).

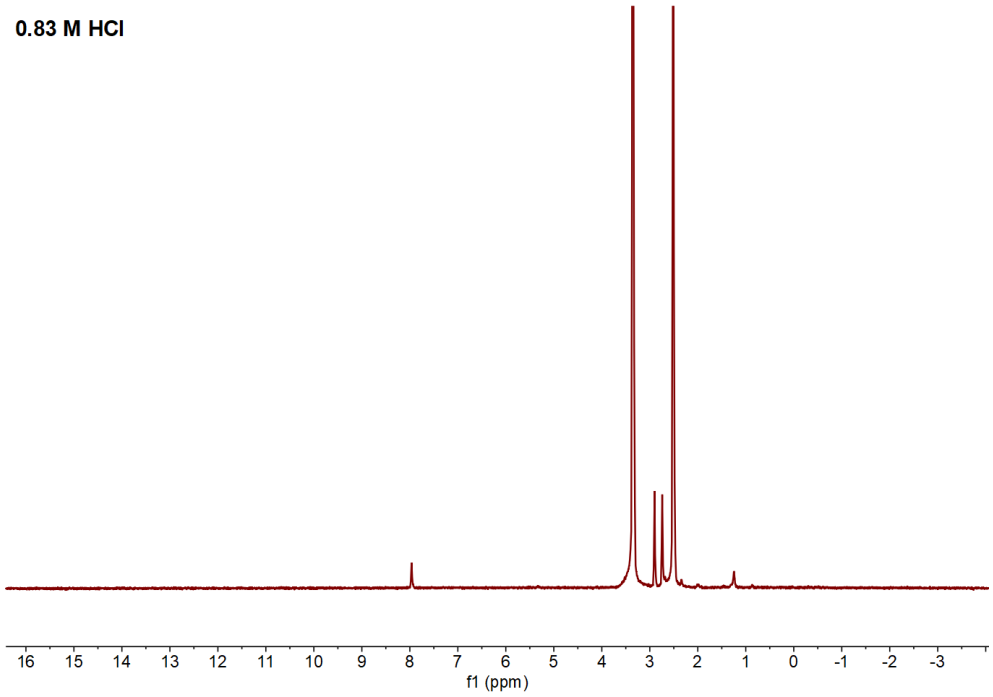


Figure S12. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.83 M HCl for 24 h (DMSO-d_6, 400 MHz).

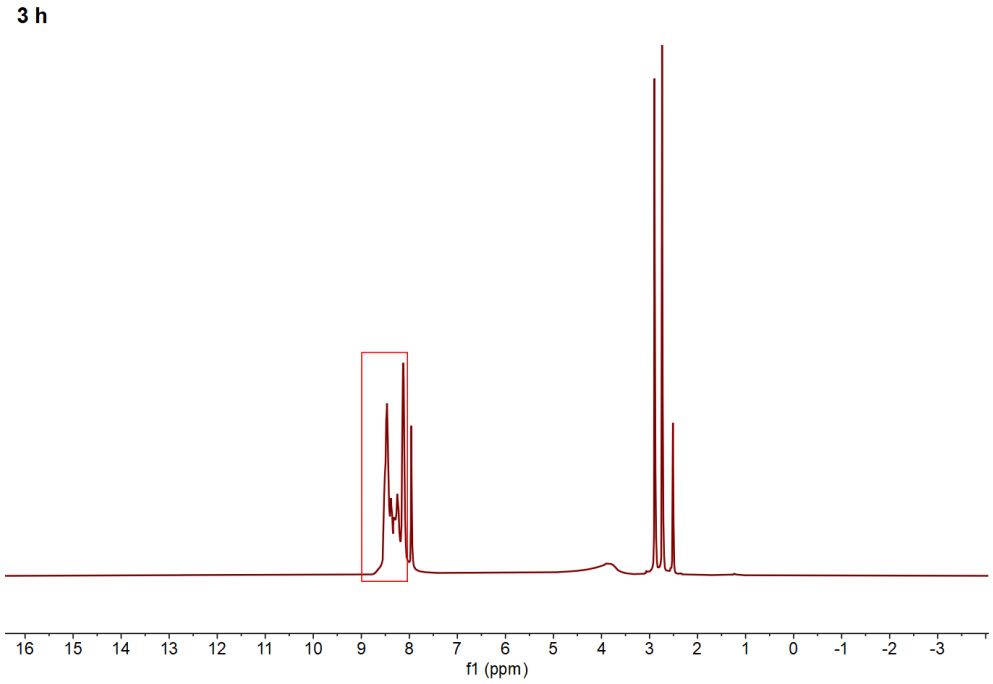


Figure S13. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.29 M HCl for 3 h (DMSO-d_6, 400 MHz).

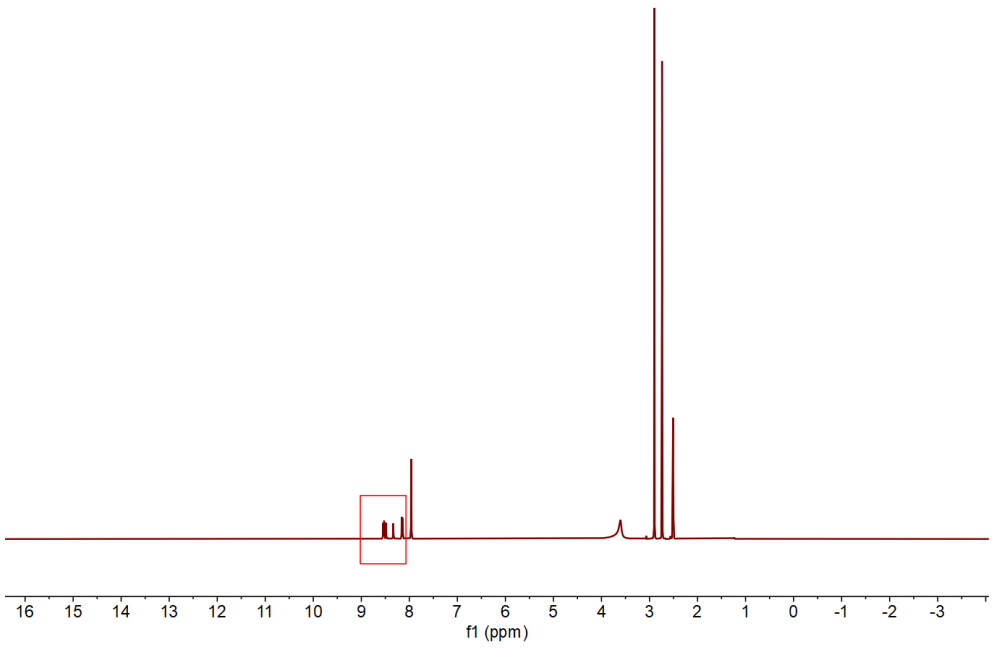


Figure S14. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.29 M HCl for 6 h (DMSO-d_6, 400 MHz).

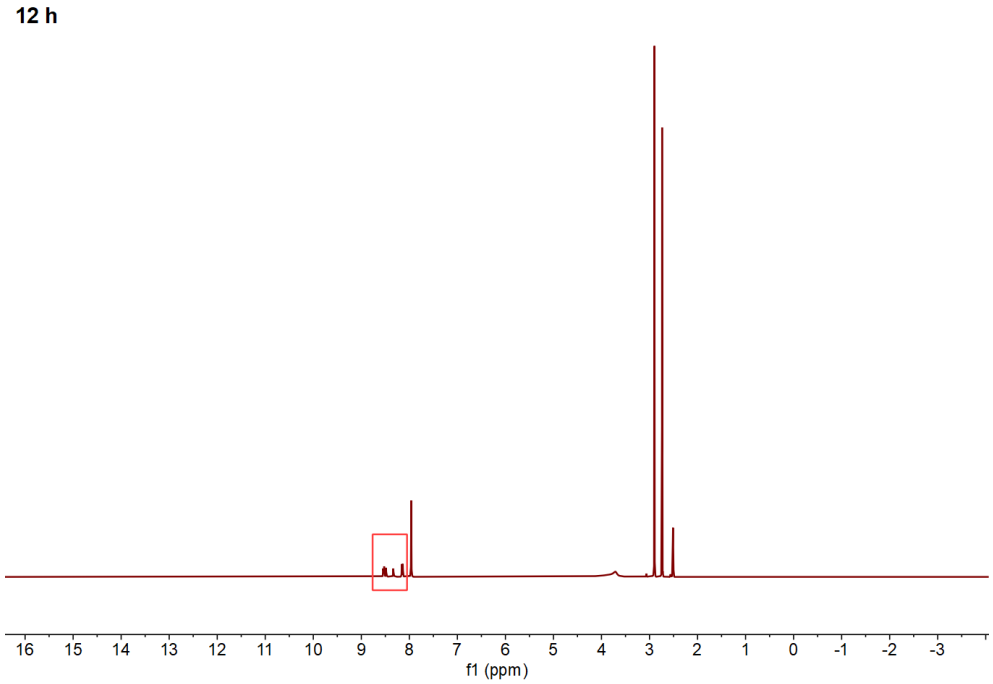


Figure S15. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.29 M HCl for 12 h (DMSO-d_6, 400 MHz).

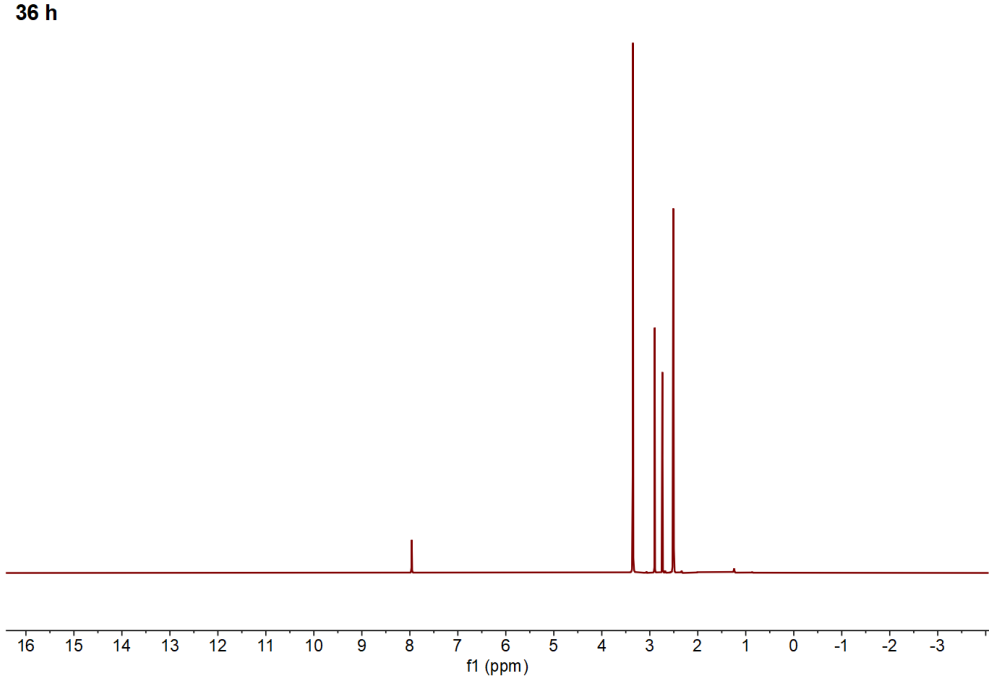


Figure S16. ^{1}H NMR spectra of the secondary supernatants derived from **Zr-PTA1** treated with 0.29 M HCl for 36 h (DMSO-d_6, 400 MHz).

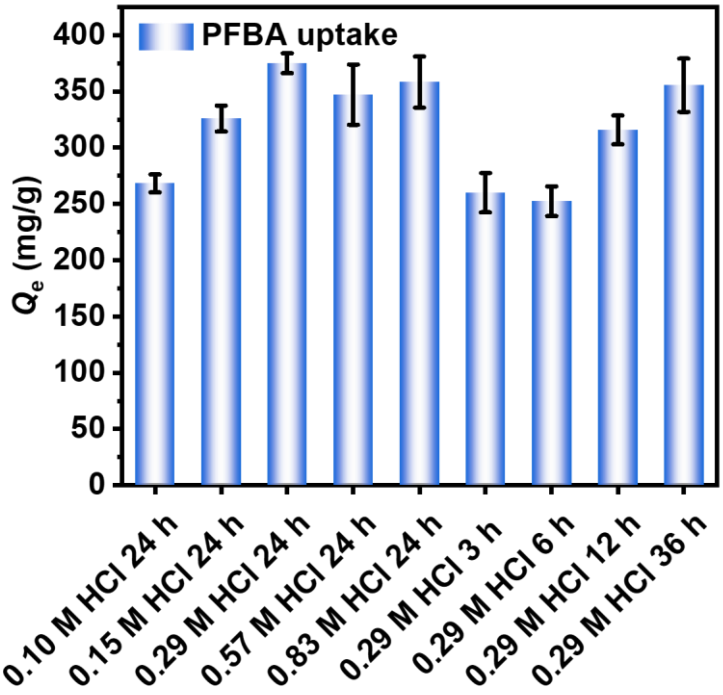


Figure S17. PFBA adsorption capacities of **Zr-PTA2** synthesized under various conditions. (Initial PFBA concentration: 400 mg/L; adsorbent dosage: 0.4 mg/mL; contact time: 24 h).

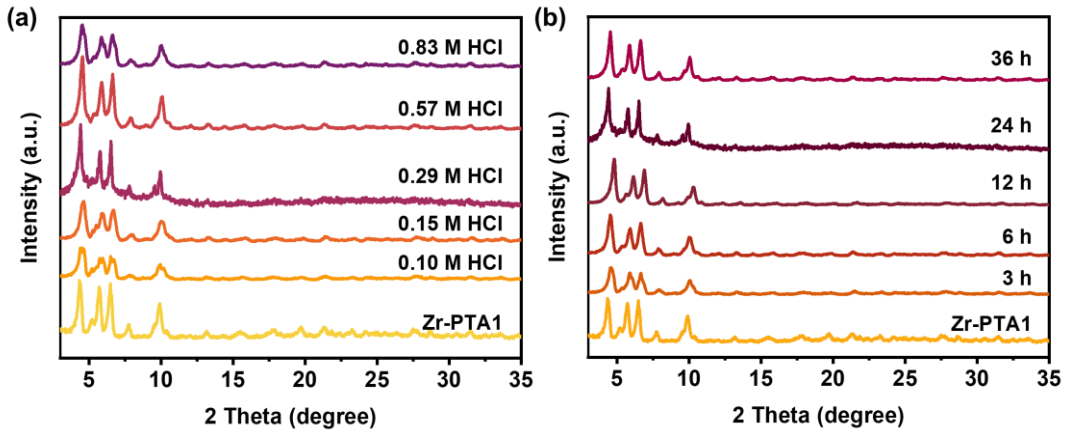


Figure S18. (a) PXRD patterns of the samples obtained under varying HCl concentrations. (b) PXRD patterns of the samples obtained under varying etching durations.

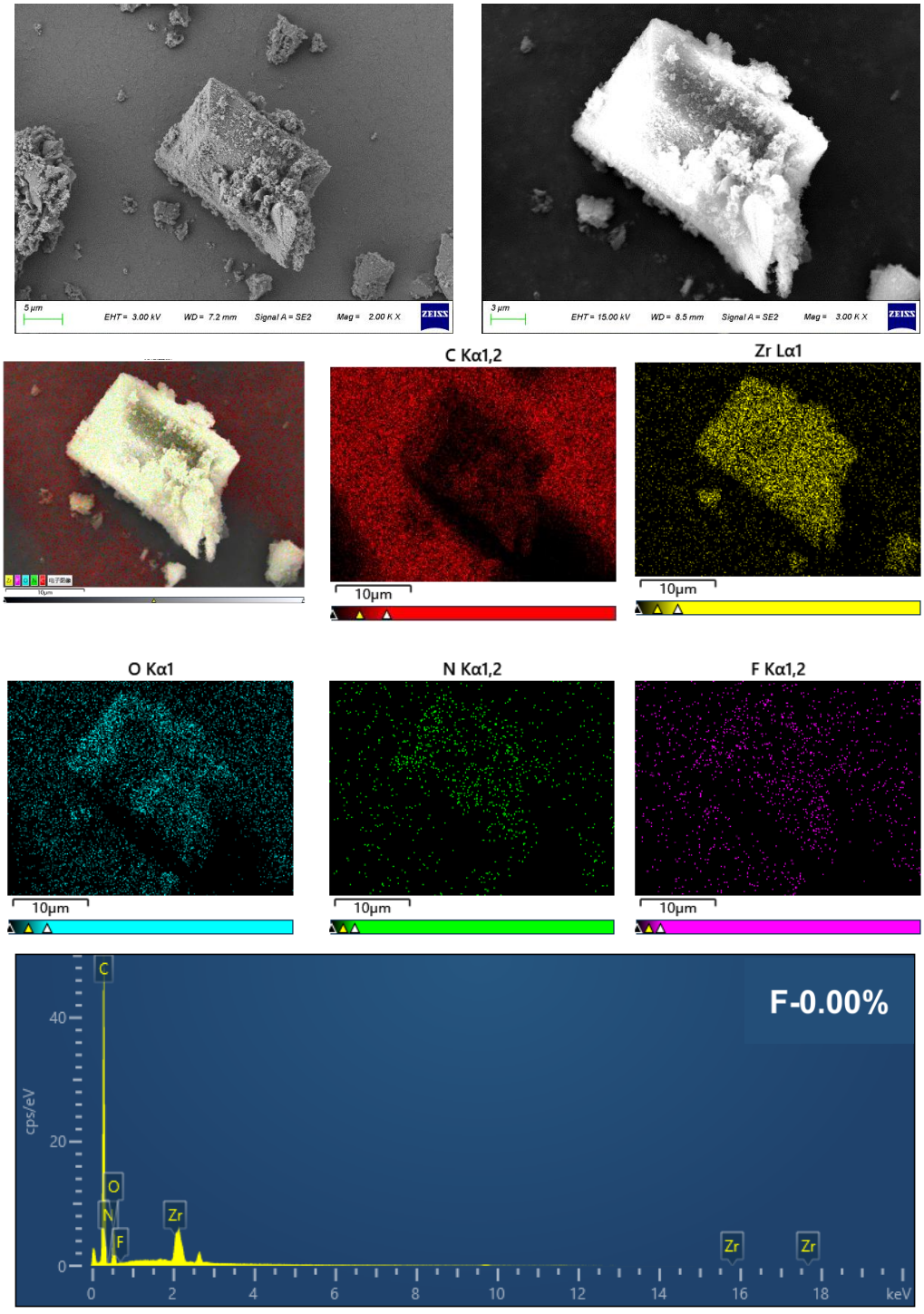


Figure S19. SEM images and EDS mapping images of **Zr-PTA2**.

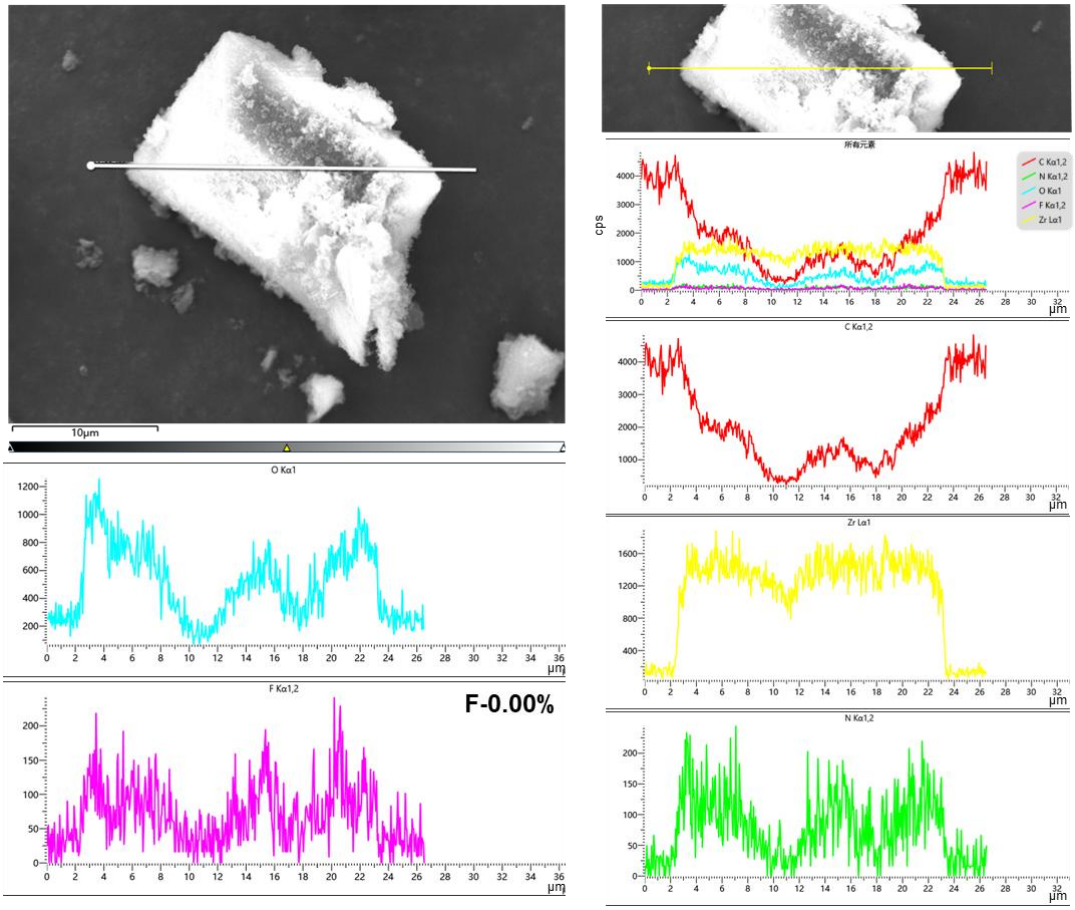


Figure S20. EDS line spectra of **Zr-PTA2**.

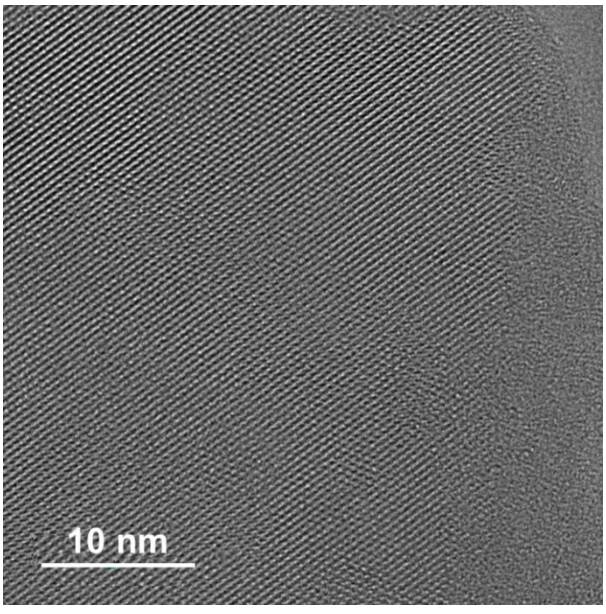


Figure S21. TEM image of **Zr-PTA1**.

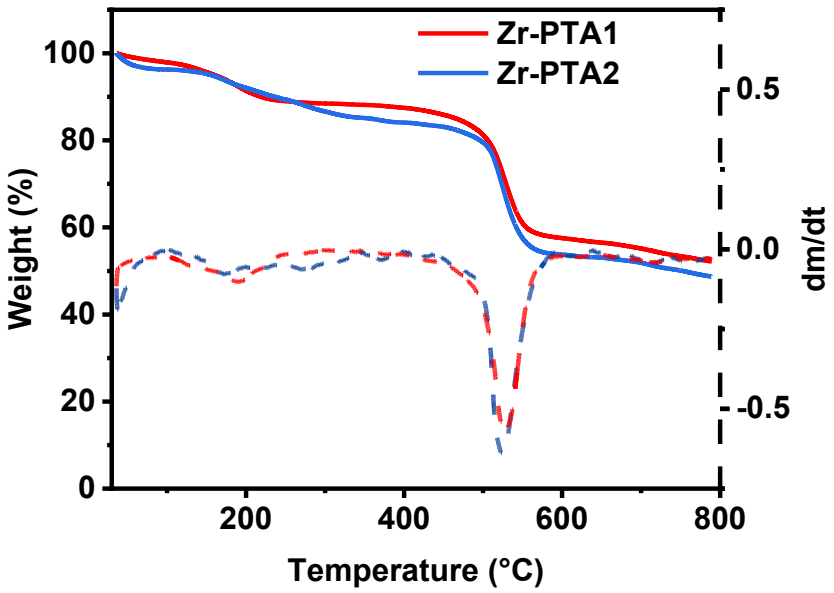


Figure S22. TGA curves of **Zr-PTA1** and **Zr-PTA2**.

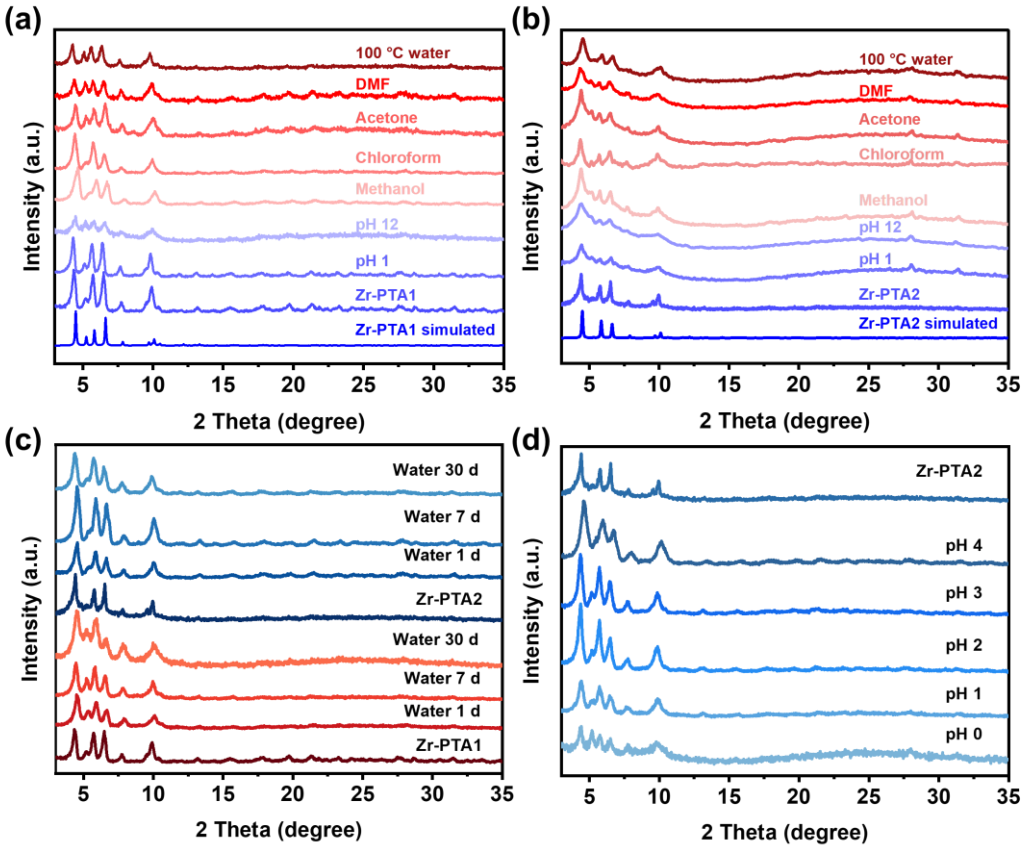


Figure S23. PXRD patterns of (a) **Zr-PTA1** and (b) **Zr-PTA2** before and after exposure to different solvents. (c) PXRD patterns of **Zr-PTA1** and **Zr-PTA2** before and after immersion in water at room temperature for 1, 7, and 30 days. (d) PXRD patterns of **Zr-PTA2** treated at different pH values (0, 1, 2, 3, and 4) for 7 days.

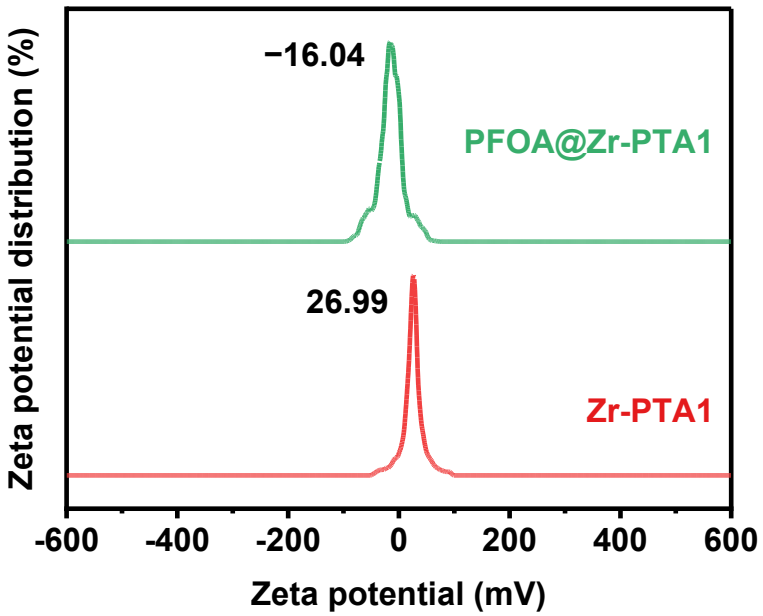


Figure S24. Zeta potential profiles of **Zr-PTA1** and PFOA@**Zr-PTA1**.

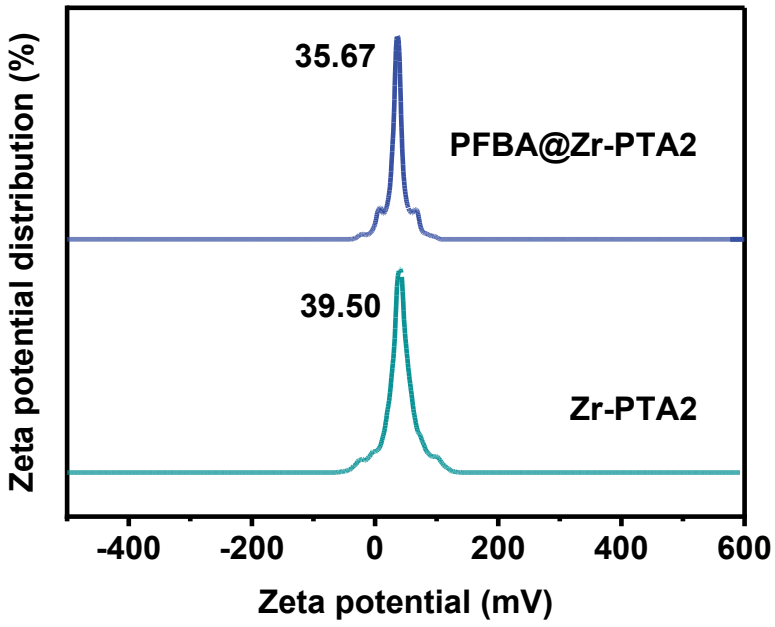


Figure S25. Zeta potential profiles of **Zr-PTA2** and PFOA@**Zr-PTA2**.

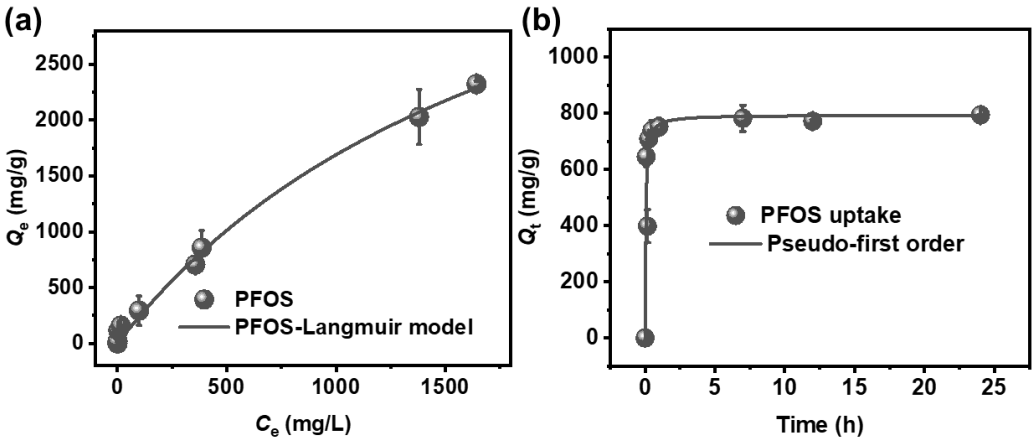


Figure S26. (a) Adsorption isotherm of PFOS on **Zr-PTA1**. (b) Pseudo-first-order kinetic model of PFOS adsorption by **Zr-PTA1**.

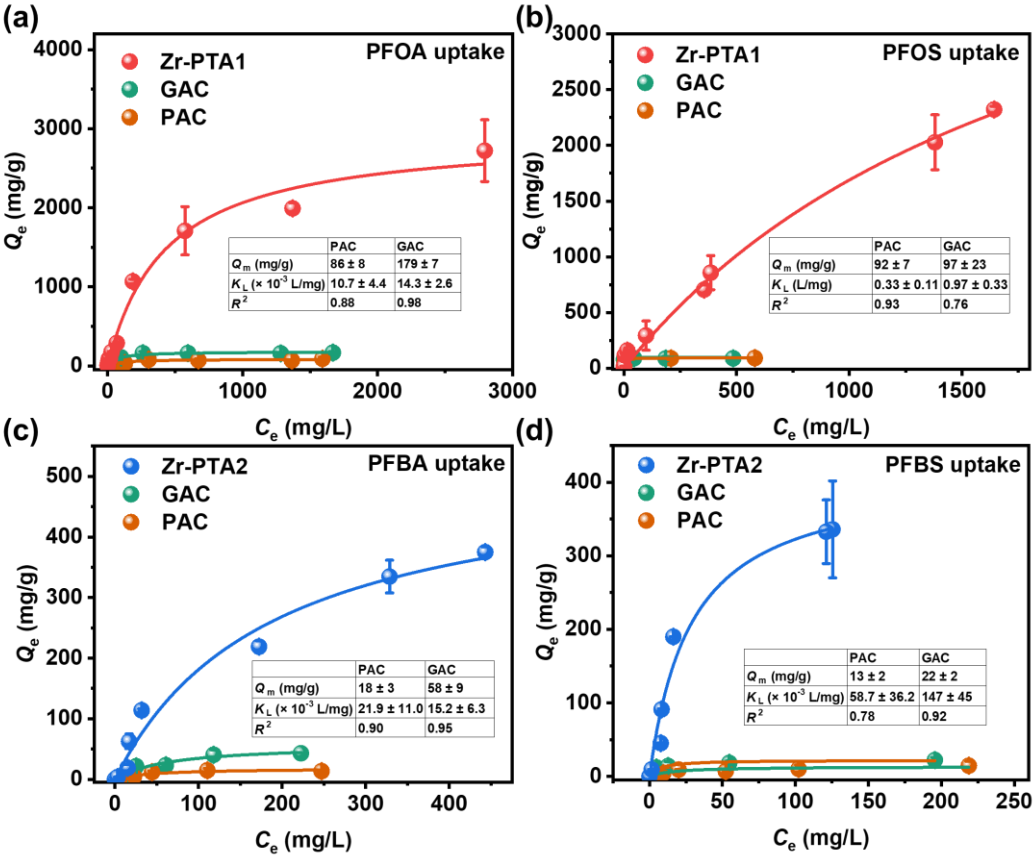


Figure S27. Adsorption isotherms of (a) PFOA, (b) PFOS, (c) PFBA, and (d) PFBS on granular activated carbon (GAC) and powdered activated carbon (PAC). The isotherms of **Zr-PTA1** and **Zr-PTA2** are included for comparison. Experimental conditions for GAC and PAC: adsorbent dosage = 0.4 mg/mL, contact time = 24 h, initial PFAS concentrations: 10–2000 mg/L for PFOA, 10–1000 mg/L for PFOS, and 5–250 mg/L for PFBA and PFBS.

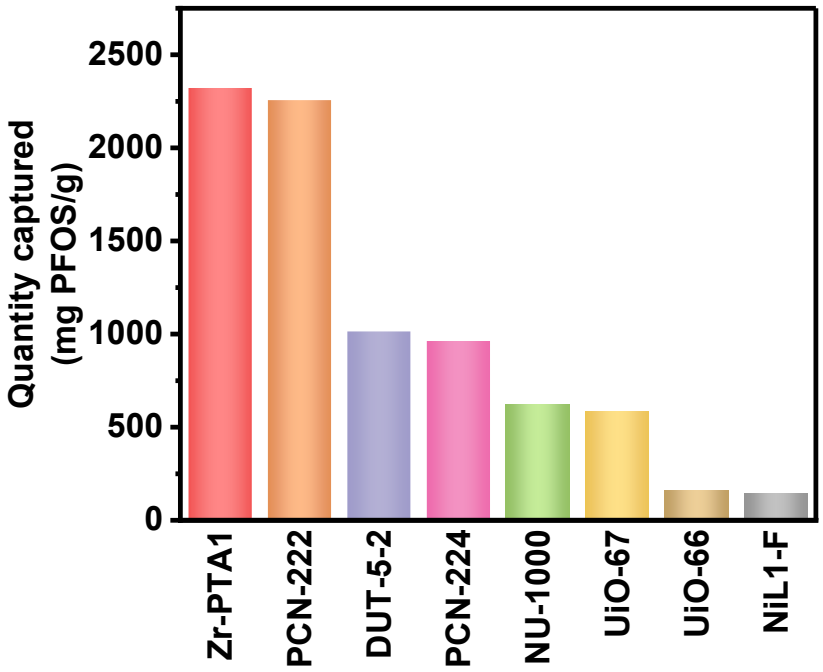


Figure S28. PFOS adsorption capacities of various reported MOF materials.



Figure S29. Photograph of the column experimental setup.

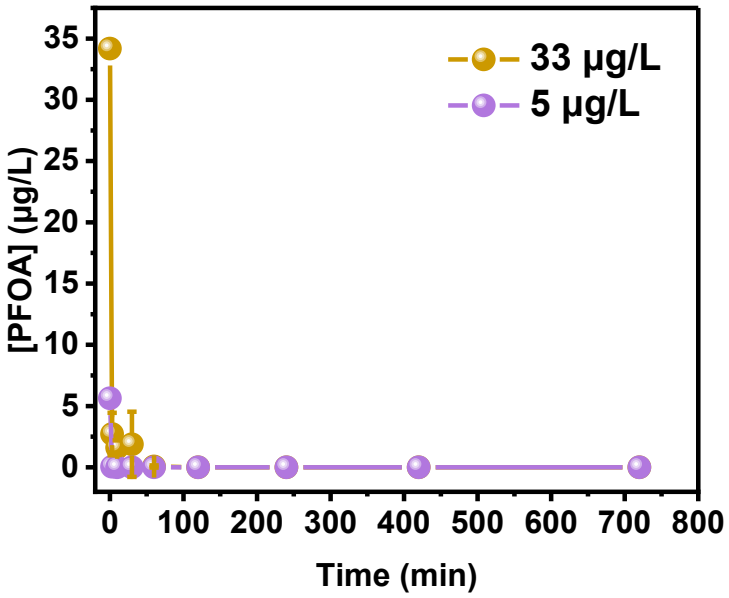


Figure S30. Zr-PTA1 adsorption performance in low-concentration PFOA.

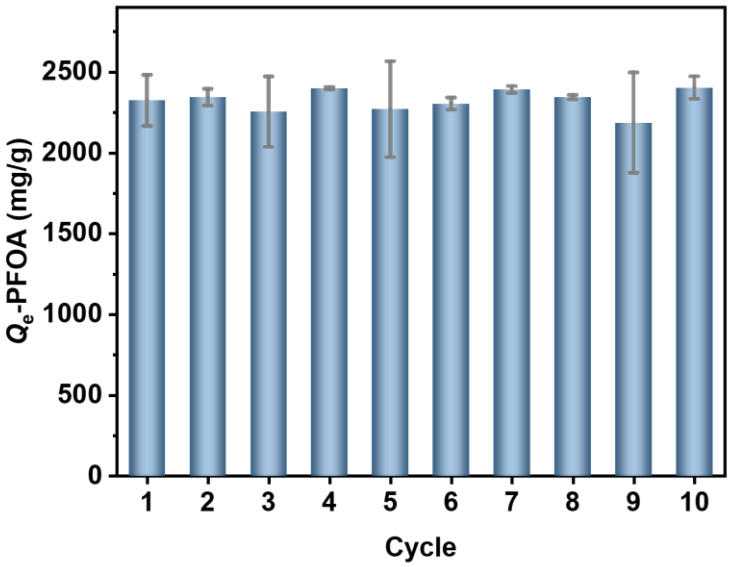


Figure S31. Reusability of **Zr-PTA1** in PFOA adsorption.

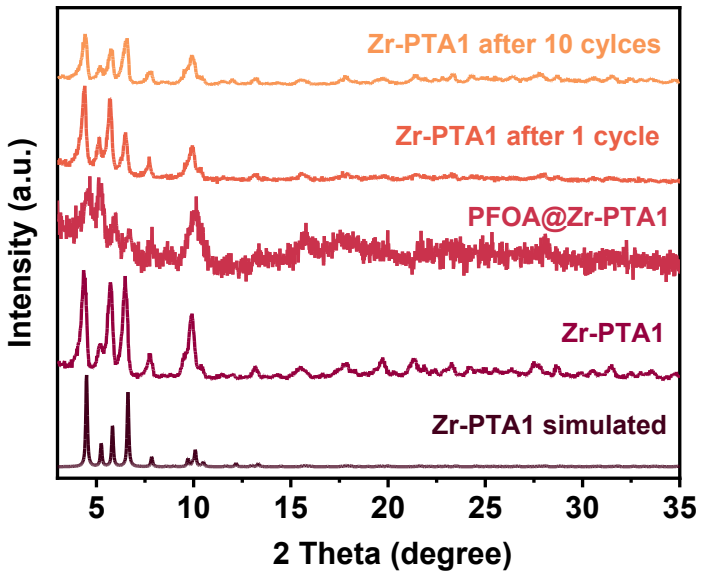


Figure S32. PXRD patterns of **Zr-PTA1** before and after PFOA adsorption and regeneration.

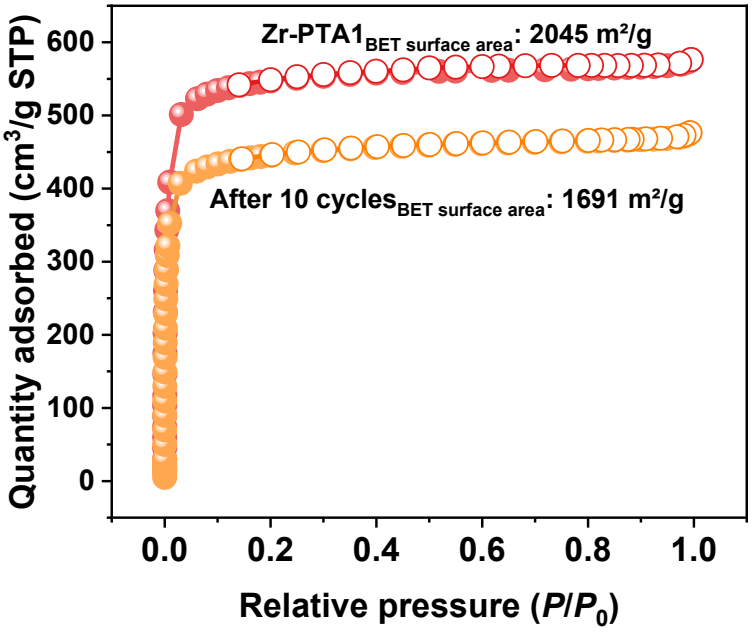


Figure S33. N_2 adsorption-desorption isotherms of **Zr-PTA1** and **Zr-PTA1** after 10 cycles.

.

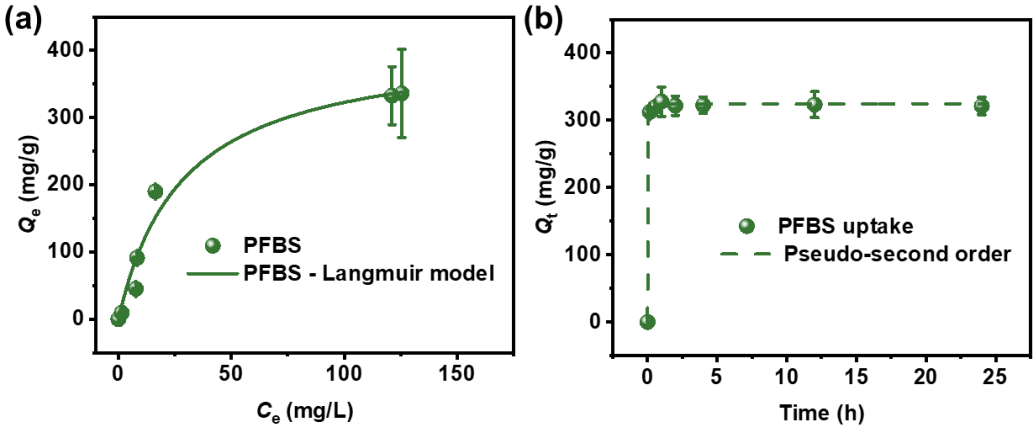


Figure S34. (a) Adsorption isotherm of PFBS on **Zr-PTA2**. (b) Pseudo-second-order kinetic model of PFBS adsorption by **Zr-PTA2**.

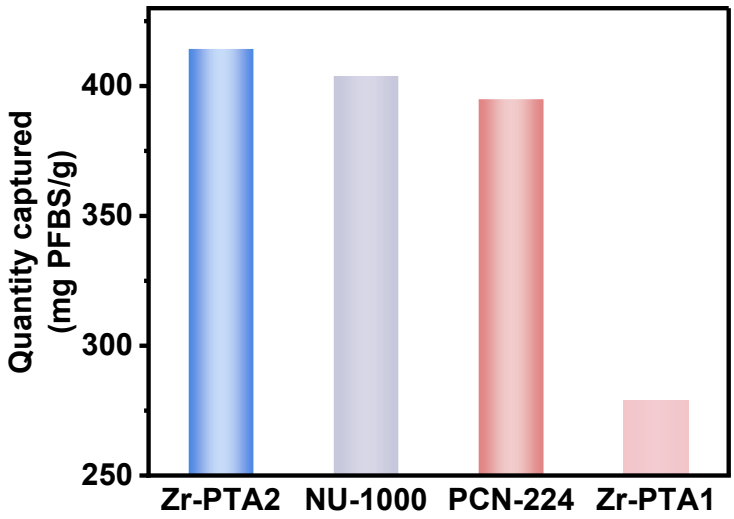


Figure S35. PFBS adsorption capacities of various reported MOF materials.

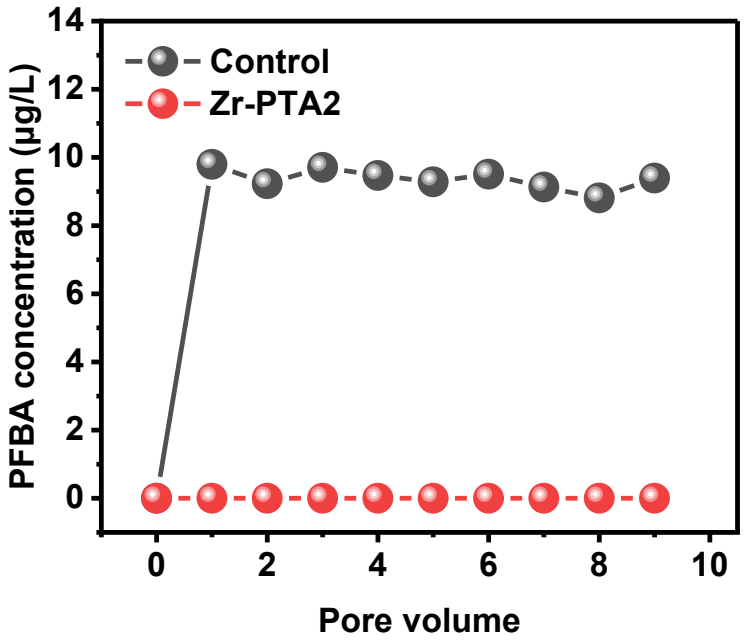


Figure S36. Breakthrough curve of PFBA on **Zr-PTA2** column.

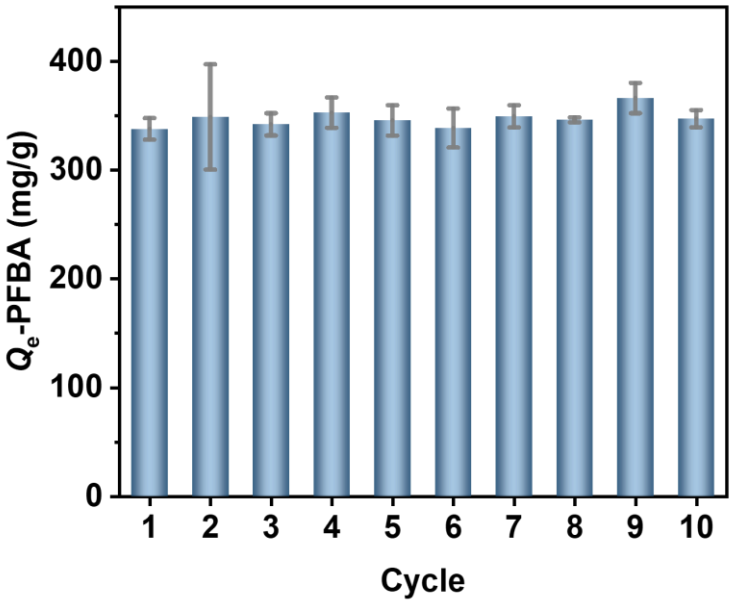


Figure S37. Reusability of **Zr-PTA2** in PFBA adsorption.

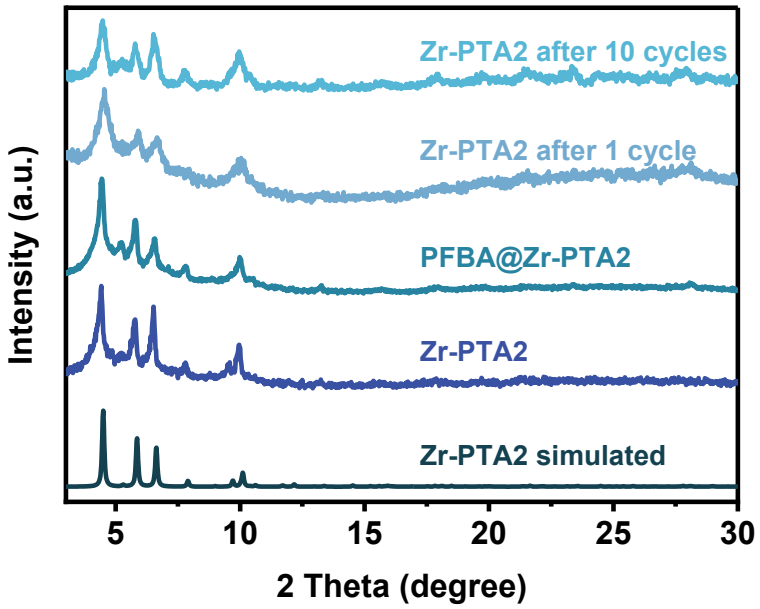


Figure S38. PXRD patterns of **Zr-PTA2** before and after PFBA adsorption.

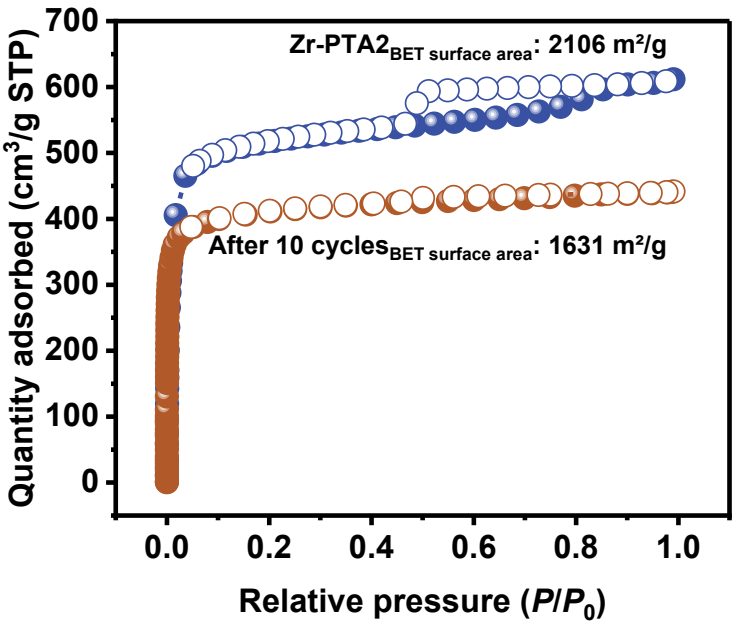


Figure S39. N_2 adsorption-desorption isotherms of **Zr-PTA2** and **Zr-PTA2** after 10 cycles.

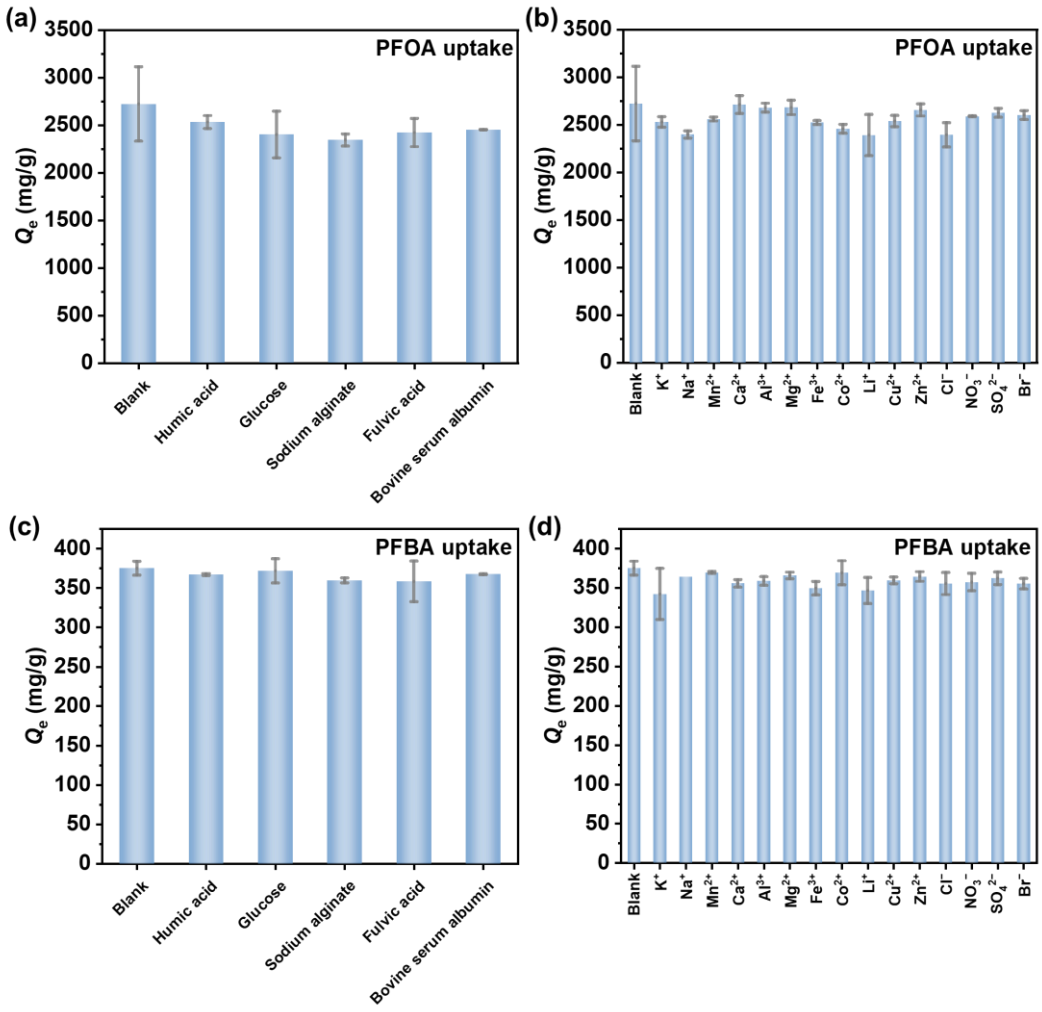


Figure S40. Equilibrium adsorption capacities of PFOA by **Zr-PTA1** in the presence of individual (a) natural organic matter (50 mg/L) and (b) inorganic ions (2 mM), and of PFBA by **Zr-PTA2** in the presence of individual (c) natural organic matter (50 mg/L) and (d) inorganic ions (2 mM).

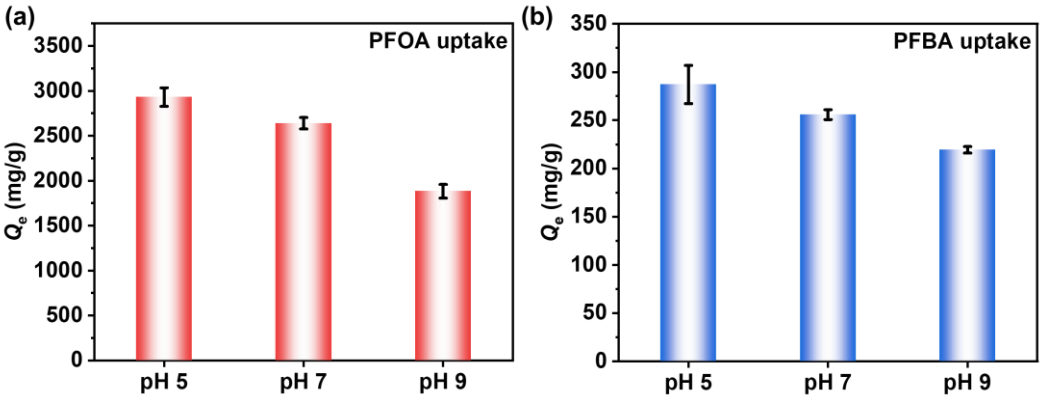


Figure S41. Equilibrium adsorption capacities of (a) PFOA by **Zr-PTA1** and (b) PFBA by **Zr-PTA2** at different initial solution pH values.

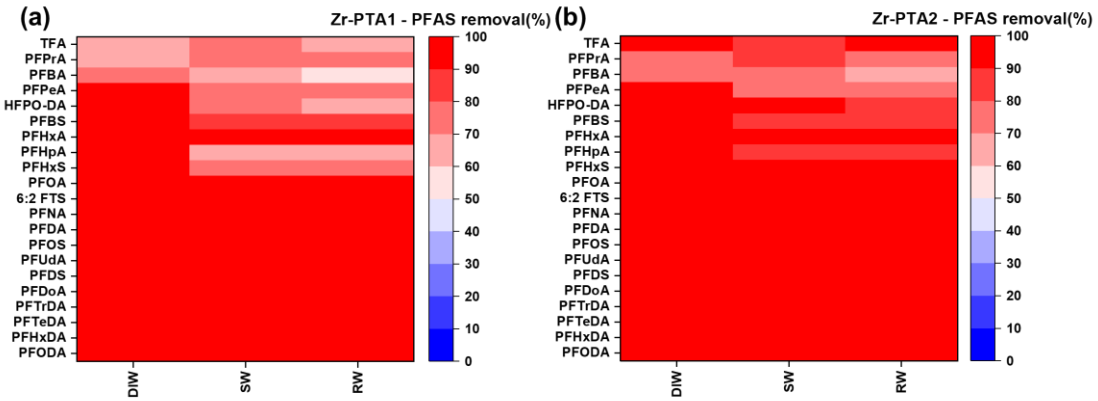


Figure S42. Removal performance of (a) **Zr-PTA1** and (b) **Zr-PTA2** for a multi-component PFAS mixture in deionized water (DIW), spring water (SW), and river water (RW). Experimental conditions: adsorbent dosage = 5 mg/mL; contact time = 24 h; initial concentrations: 100 μg/L for TFA and 1 μg/L for each of the other PFAS species.

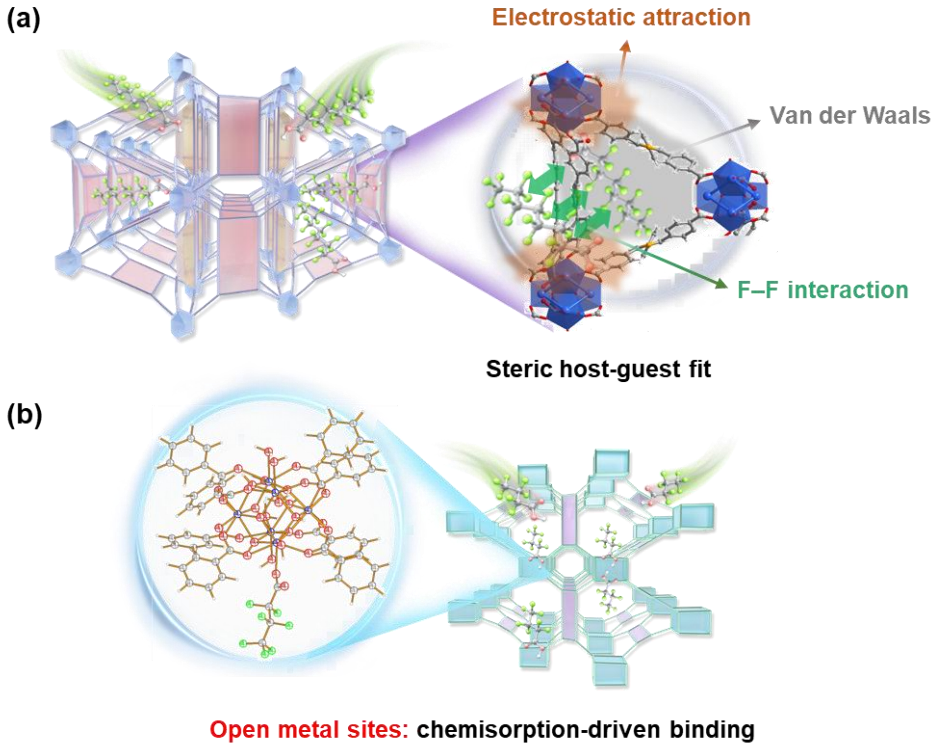


Figure S43. (a) Schematic illustration of the adsorption mechanism of PFOA by **Zr-PTA1**. (b) Schematic illustration of the adsorption mechanism of PFBA by **Zr-PTA2**.

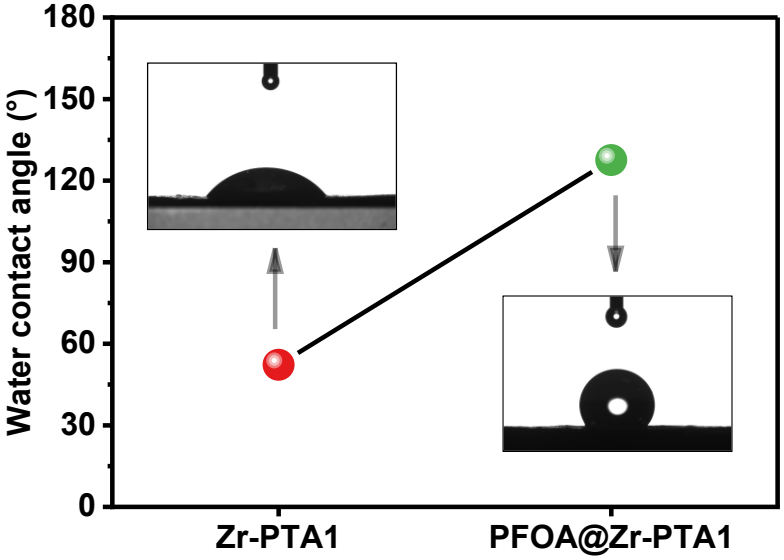


Figure S44. Static water contact angles of **Zr-PTA1** and PFOA@**Zr-PTA1**.

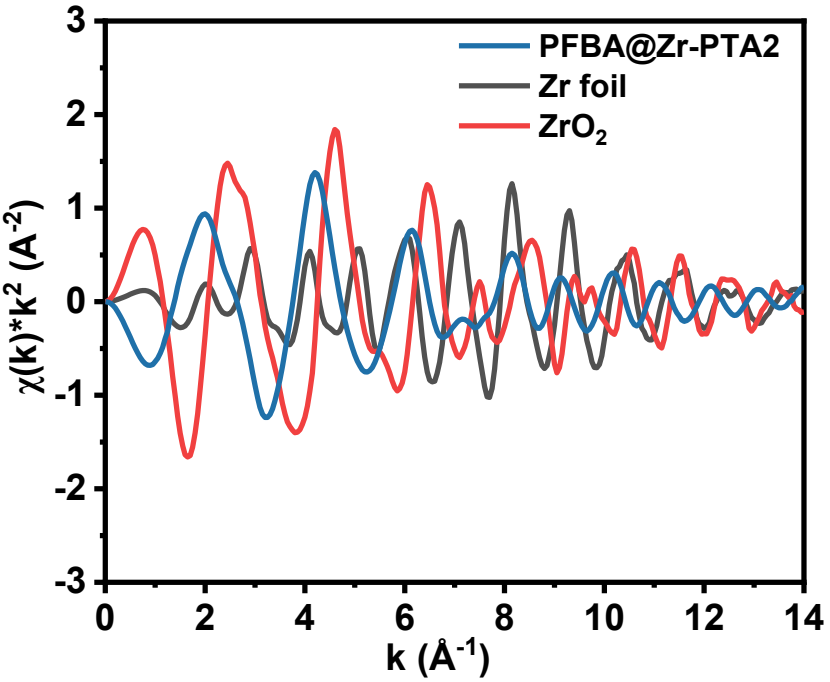


Figure S45. Zr K-edge extended XANES oscillation functions $k^3\chi(k)$ for PFBA@**Zr-PTA2**.

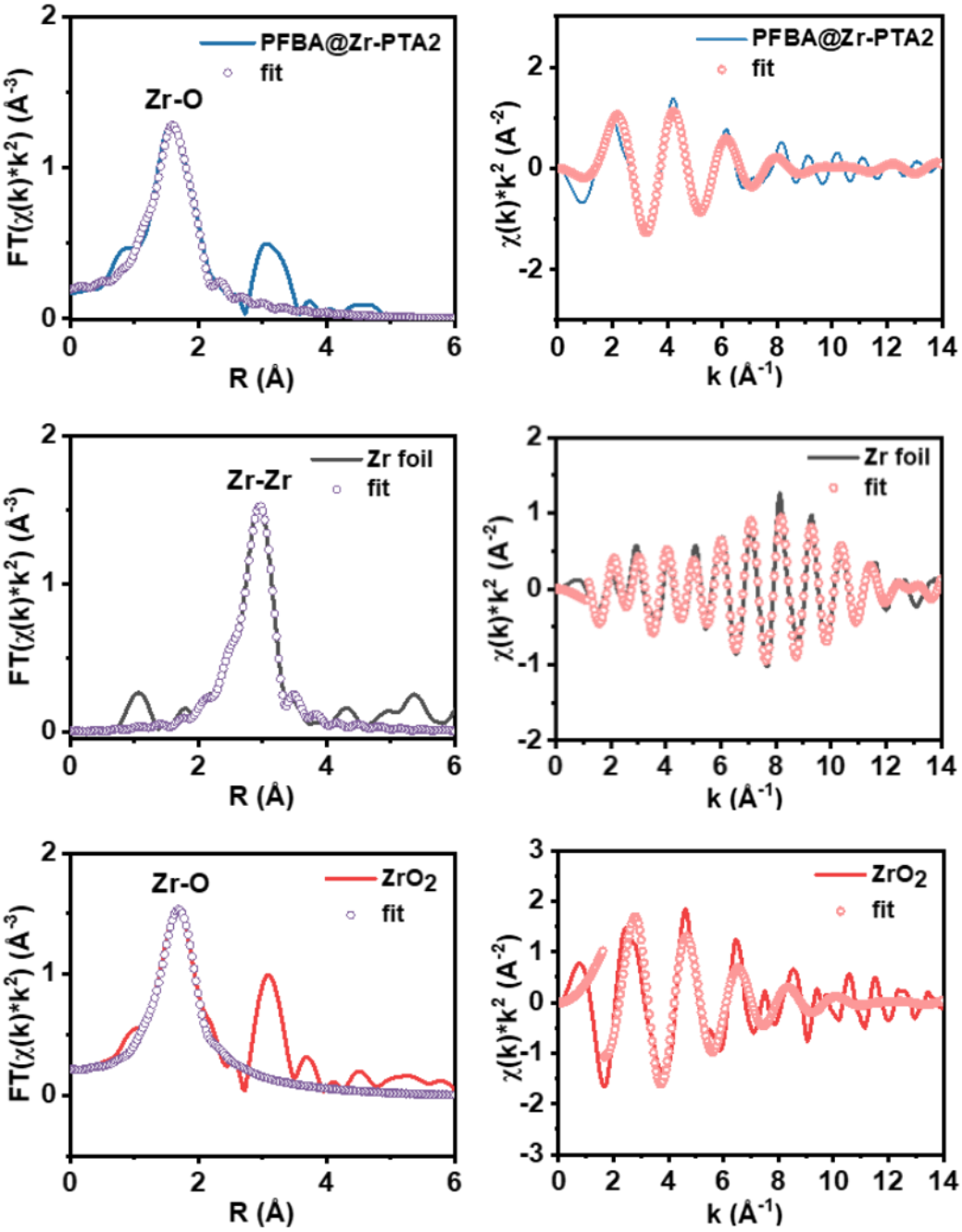


Figure S46. Zr K-edge EXAFS (line) and the curve fit (points) for PFBA@**Zr-PTA2**, Zr foil, and ZrO_2 shown in R-space (Fourier transform magnitude and real component) and k^3-weighted k-space.

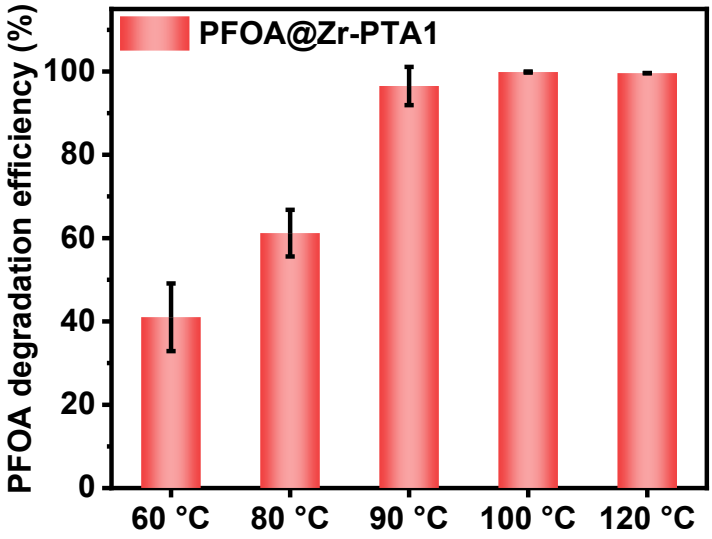


Figure S47. Effect of reaction temperature on PFOA degradation. Experimental conditions: PFOA@**Zr-PTA1** with a PFOA loading of ~500 mg/L, 15 equiv NaOH, DMSO/H_2O = 8:1 v/v, *t* = 45 h.

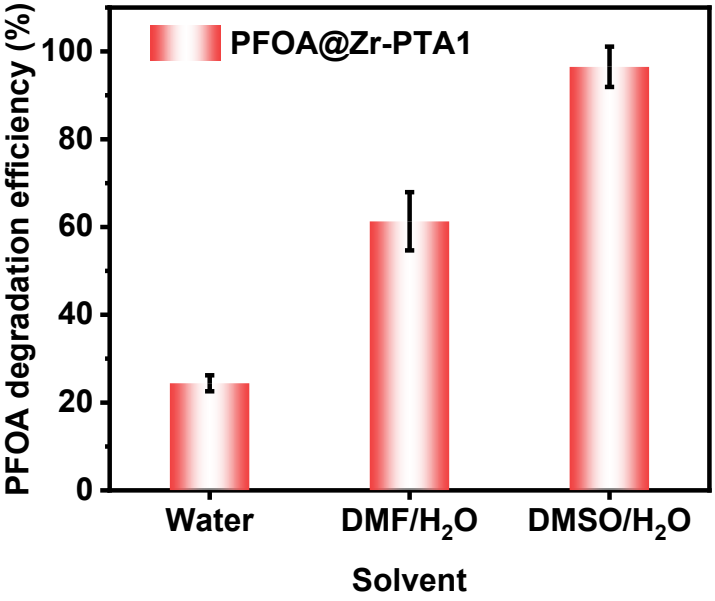


Figure S48. Effect of solvent on the thermocatalytic degradation of PFOA by **Zr-PTA1**. Experimental conditions: PFOA@**Zr-PTA1** with a PFOA loading of ~500 mg/L, 15 equiv NaOH, solvent ratio = 8:1 v/v, *T* = 90 °C, *t* = 45 h.

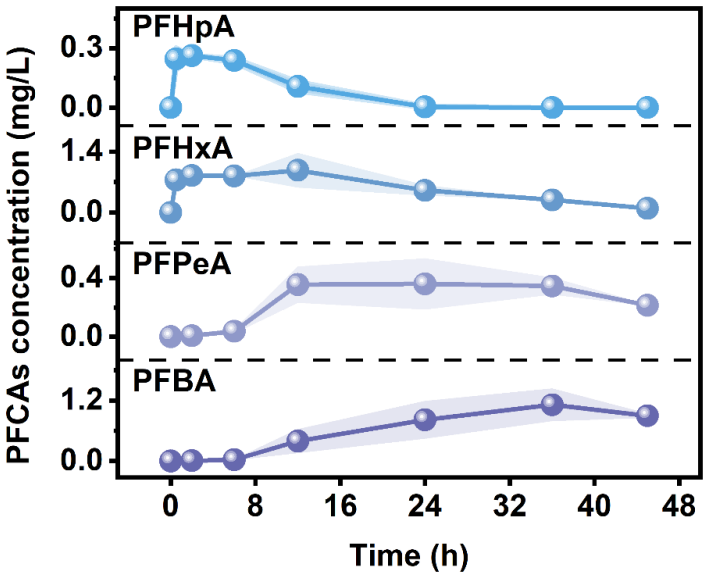


Figure S49. Concentration changes of C4–C7 perfluorinated carboxylic acids during PFOA degradation.

Supplementary tables

Table S1. Crystal data and structure refinement for **Zr-PTA1**.

Identification code	**Zr-PTA1**
Empirical formula	$C_{66}H_{42}N_4O_{18}Zr_3$
Formula weight	1452.69
Temperature/K	100.00
Crystal system	tetragonal
Space group	$I4_1/amd$
a/Å	20.5378(4)
b/Å	20.5378(4)
c/Å	67.313(2)
α/°	90
β/°	90
γ/°	90
Volume/Å^3	28392.5(14)
Z	8
ρ_{calc} g/cm^3	0.680
μ/mm^{-1}	2.058
F(000)	5840.0
Crystal size/mm^3	0.13 × 0.11 × 0.1
Radiation	Cu Kα (λ = 1.54178)
2*θ* range for data collection/°	4.498 to 142.016
Index ranges	–25 ≤ h ≤ 25, –25 ≤ k ≤ 25, –82 ≤ l ≤ 81
Reflections collected	301015
Independent reflections	7227 [R_{int} = 0.0783, R_{sigma} = 0.0183]
Data/restraints/parameters	7227/538/380
Goodness-of-fit on F^2	1.078
Final R indexes [I>=2σ (I)]	R_1 = 0.0518, wR_2 = 0.1609
Final R indexes [all data]	R_1 = 0.0595, wR_2 = 0.1726
Largest diff. peak/hole / e Å^{-3}	0.59/–0.92

Table S2. Crystal data and structure refinement for **Zr-PTA2**.

Identification code	**Zr-PTA2**
Empirical formula	$C_{48}H_{30}N_2O_{16}Zr_3$
Formula weight	1164.44
Temperature/K	293(2)
Crystal system	Tetragonal
Space group	$I4_1/amd$
a/Å	20.55(10)
b/Å	20.55(10)
c/Å	66.6(3)
α/°	90
β/°	90
γ/°	90
Volume/Å^3	28125(299)
Z	8
ρ_{calc} g/cm^3	0.550
μ/mm^{-1}	0.000
F(000)	1670.0
Radiation	electron (λ = 0.0251)
2*θ* range for data collection/°	0.074 to 1.798
Index ranges	−25 ≤ h ≤ 25, –25 ≤ k ≤ 25, –82 ≤ l ≤ 82
Reflections collected	49940
Independent reflections	7527 [R_{int} = 0.5495, R_{sigma} = 0.3306]
Data/restraints/parameters	7527/1/76
Goodness-of-fit on F^2	0.888
Final R indexes [I>=2σ (I)]	R_1 = 0.1676, wR_2 = 0.3891
Final R indexes [all data]	R_1 = 0.2433, wR_2 = 0.4396
Largest diff. peak/hole / e Å^{-3}	0.22/–0.28

Table S3. Fitting parameters of the Langmuir model for the adsorption of PFOA and PFOS by **Zr-PTA1**.

Model	Parameter	PFOA	PFOS
Langmuir	Q_m (mg/g)	2945 ± 173	5093 ± 796
	K_L (× 10^{-3} L/mg)	2.03 ± 0.44	0.50 ± 0.13
	R^2	0.98	0.99

Table S4. Adsorption capacities of various MOFs and activated carbon materials for PFOA, PFOS, PFBA, and PFBS.

Adsorbent	PFOA concentration mg/L	(Maximum) Adsorption capacity mg PFOA/g	PFOS concentration mg/L	(Maximum) Adsorption capacity mg PFOS/g	PFBA concentration mg/L	(Maximum) Adsorption capacity mg PFBA/g	PFBS concentration mg/L	(Maximum) Adsorption capacity mg PFBS/g	Reference
Zr-PTA1	**10–4000**	**2945**	**2000**	**2322**	**800**	**294**	**800**	**279**	**This work**
Zr-PTA2	**—**	**—**	**—**	**—**	**400**	**375**	**10–200**	**414**	**This work**
TFA-MOF-808	0–800	2496	—	—	100	311	—	—	[13]
MOF-808	0–800	1581	—	—	100	232	—	—	[13]
PCN-999	1000	1089	—	—	1000	236	—	—	[14]
MIL-101(Cr)-QDMEN	0–400	783	—	—	—	—	—	—	[15]
UIO-67	1000	743	1000	585	—	—	—	—	[16]
PCN-1003	1000	642	—	—	—	—	—	—	[17]
PCN-1002	2000	632	—	—	—	—	—	—	[18]
NU-1000	100	507	100	622	100	274	100	404	[19]
MIL-101(Cr)-DMEN	0–400	493	—	—	—	—	—	—	[15]
MIL-101-Fe	50–1000	490	—	—	—	—	—	—	[20]
TFA-DUT-67	500	490	—	—	—	—	—	—	[13]
DUT-5-2	10–140	474	10–140	1015	—	—	—	—	[21]
Zn(aIM+bIM)	1–100	439	—	—	100	245	—	—	[22]
F-MOF	—	420	—	—	—	—	—	—	[23]
UiO-66	1000	388	1000	160	—	—	—	—	[16]
DUT-67	500	383	—	—	—	—	—	—	[13]
La-MOF	1–100	364	—	—	—	—	—	—	[24]
Zn(mIM+bIM)	1–100	343	—	—	100	210	—	—	[22]
Zr-MOF	1–100	340	—	—	—	—	—	—	[24]
NiL1-F	10–100	269	—	142	—	—	—	—	[25]
PCN-1001	2000	174	—	—	—	—	—	—	[18]
PCN-222	—	—	500	2257	—	—	—	—	[26]
PCN-224	—	—	50–500	963	—	—	50–500	395	[27]
GAC	50	248	50	334	—	—	—	—	[28]
GAC	~0.8	137	~1	211	~0.4	49	~0.5	75	[29]

GAC	20–250	170	—	199	—	—	—	—	[30]
CNTs	50	166	50	215	—	—	—	—	[28]
graphene	50	329	50	528	—	—	—	—	[28]
Fresh CAC	0.10–6.21	81	—	—	0.05–3.21	25	0.08–4.5	18	[31]
H-CAC	0.10–6.21	87	—	—	0.05–3.21	69	0.08–4.5	26	[31]
C-IS	0.2	~2	0.003–0.015	~3.5	—	—	—	—	[32]
CAC	0.005-1	5	—	24	—	0.2	—	3	[33]
CS/GAC	—	298	—	—	—	—	—	—	[34]
PAC	20–250	292	20–250	559	—	—	—	—	[30]
Cetyltrimethylammonium bromide-functionalized graphene oxide	25-200	714	—	—	—	—	—	—	[35]
Activated carbon	0.125-1.0	79	0.125-1.0	75	—	—	—	—	[36]
Chemically activated tassel	10-100	239	—	204	—	—	—	—	[37]

GAC: granular activated carbon; CNT: carbon nanotubes; CAC: colloidal activated carbon; H-CAC: hydrogen peroxide-modified colloidal activated carbon; C-IS: inorganic silicon modified activated carbon; CS/GAC: chitosan-loaded granular activated carbon; PAC: powdered activated carbon.

Table S5. Kinetic parameters of pseudo-first-order model fittings for PFOA and PFOS adsorption by **Zr-PTA1**.

Kinetics model	Parameter	PFOA	PFOS
Pseudo-first order	Q_1 (mg/g)	2203 ± 116	751 ± 51
	k_1 (h^{-1})	3.44 ± 0.78	11.4 ± 4.2
	R_1^2	0.89	0.83

Table S6. Fitting parameters of the Langmuir model for the adsorption of PFBA and PFBS by **Zr-PTA2**.

Model	Parameter	PFBA	PFBS
Langmuir model	Q_m (mg/g)	519 ± 79	414 ± 41
	K_L (× 10^{-3} L/mg)	5.37 ± 2.06	35.2 ± 10.9
	R^2	0.98	0.97

Table S7. Kinetic parameters of pseudo-second-order model fittings for PFBA and PFBS sorption by **Zr-PTA2**.

Kinetics model	Parameter	PFBA	PFBS
Pseudo-second order	Q_2 (mg/g)	285 ± 8	324 ± 3
	k_2 (× 10^{-3} g/(mg·h))	67.5 ± 13.1	397 ± 189
	R_2^2	0.99	1.00

Table S8. Elemental analysis of spring water (SW) and river water (RW) determined by ICP-MS.

Element	Content (μg/L)	Element	Content (μg/L)
SW		RW	
Ca	61487	Ca	129668
Na	22933	Na	112418
Mg	21798	S	83684
Si	9862	Mg	41029
S	2600	Si	3879
K	818	K	3442
Sr	466	Sr	1518
B	88	B	222
P	40	Ba	63
Ba	36	P	19
Cr	27	Li	12
Li	2.2	Sb	3.1
V	2.1	Al	2.9
Zn	2.0	Zn	2.0
As	1.1	Mo	1.8
Se	0.93	As	1.4
Mo	0.93	Mn	1.0
Pb	0.45	V	0.98
Ni	0.44	Cu	0.92
Rb	0.18	Rb	0.84
Cs	0.13	Cr	0.84
Cu	0.11	Ni	0.70
Tb	0.097	Pb	0.30
Zr	0.089	Se	0.29
Cd	0.047	Cs	0.14
Mn	0.036	Zr	0.12
Ru	0.035	Co	0.071
Ir	0.023	Cd	0.060
Ge	0.021	Ge	0.028
Co	0.016	La	0.026
In	0.014	Ga	0.020
Nb	0.0065	Tb	0.018
La	0.0055	Ir	0.012
Sb	0.0048	Nd	0.012
Bi	0.0043	Ru	0.0095
Hf	0.0023	Nb	0.0083
Hg	0.0018	Bi	0.0071
Au	0.0013	Gd	0.0068
Nd	0.00091	In	0.0046
Dy	0.00084	Dy	0.0038
Pd	0.00049	Pr	0.0036
Eu	0.00022	Ag	0.0034
		Lu	0.0033
		Pd	0.0032

Hf	0.0032
Hg	0.0024
Au	0.0024
Tl	0.0020
Ho	0.0018
Eu	0.0012
Tm	0.00092

Table S9. EXAFS fitting parameters at the Zr K-edge for various samples.

Sample	Shell	CN[a]	R (Å)[b]	σ^2 (Å^2)[c]	ΔE_0 (eV)[d]	K-range (Å^{-1})	R-range (Å^{-1})	R factor
Zr foil	Zr-Zr_1	6[e]	3.14 ± 0.00	0.0020 ± 0.0013	5.18 ± 0.47	2.7−12.2	1.4−3.5	0.0064
	Zr-Zr_2	6[e]	3.26 ± 0.01	0.0021 ± 0.0013				
ZrO_2	Zr-O_1	4[e]	2.10 ± 0.01	0.0030	10.32 ± 1.53	3.0−11.7	1.1−2.4	0.015
	Zr-O_2	4[e]	2.23 ± 0.02					
	Zr-O_2	3.8 ± 0.6	2.30 ± 0.02					
PFBA@**Zr-PTA2**	Zr-O_1	3.0 ± 0.4	2.12 ± 0.01	0.0020	−0.22 ± 1.06	2.6−12.4	1.0−2.4	0.011
	Zr-O_2	4.9 ± 0.4	2.27 ± 0.01	0.0040				

[a] CN, coordination number; [b] R, distance between absorber and backscatter atoms; [c] σ^2, Debye-Waller factor to account for both thermal and structural disorders; [d] ΔE_0, inner potential correction; R factor indicates the goodness of the fit. S_0^2 was fixed to 0.85 for Zr foil, according to the experimental EXAFS fit of Zr foil by fixing CN as the known crystallographic value. [e] fixed value. A reasonable range of EXAFS fitting parameters: $0.700 < S_0^2 < 1.000$; CN > 0; σ^2Å^2 > 0; $|\Delta E_0| < 15$ eV; R factor < 0.02.

Table S10. Solvent gradient program.

Time (min)	Flowrate (mL/min)	%A	%B
Initial	0.3	80.0	20.0
0.5	0.3	80.0	20.0
3.0	0.3	10.0	90.0
4.0	0.3	10.0	90.0
4.5	0.3	0.0	100.0
5.0	0.3	90.0	10.0

Table S11. Analytical details for PFAS.

Compound	Parent (m/z)	Daughter (m/z)	Cone (V)	Collision (V)
PFOA	413	369	15	10
PFOS	499	99	10	35
PFBA	213	169	18	8
PFBS	299	99	2	24
PFPeA	263	219	19	8
PFHxA	313	269	12	9
PFHpA	363	319	18	10

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